Observer-Relative Soft and Hard FTL in the Structural-Physical Framework

Theory manuscript with explicit audit boundary

Abstract

This manuscript studies faster-than-light transport only in observer-relative form, relative to a chosen luminal reference sector inside an admissible structural-physical regime. Its first main result is a closed soft theorem package consisting of branch-locked witness structure, explicit binary readout, finite and hierarchical scanner organization, detector-memory operationalization, and robustness of remote-scene approximation.

Its second main result is a hard theorem chain built around explicit branch-transfer carrier families, projected-pole survival for the explicit family, identity-preserving carrier cycles, composite binding criteria, explicit bound-transport prototypes, and quantitative one-pole and composite robustness margins. On top of this explicit chain, the manuscript proves the general hard-closure theorems H-C1 through H-C5 from fixed perturbation norms, positivity margins, openness lemmas, continuity estimates, and projection-compatibility hypotheses. The manuscript also states the integrated soft-hard mission layer and the pre-engineering boundary without conflating theorem-level closure with prototype claims. Failure modes and non-claims are included as part of the final audit boundary.

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1 Introduction and Claim Boundary

The FTL program does not treat faster-than-light transport as a primitive law of the underlying theory. In the parent framework, velocity appears only after the choice of an observer window, a spatial projection, and a temporal slicing. For that reason, every FTL statement in this manuscript is observer-relative from the start: a sector is called superluminal only relative to a chosen luminal reference sector inside one fixed effective observational regime. The manuscript therefore forbids the informal reading in which "FTL" means a universal violation of causality or a direct contradiction of the fundamental structural order.

This observer-relative restriction determines the writing protocol of the entire document. We must distinguish three layers that are often conflated in speculative transport writing but are sharply separated here. First, there is the structural layer, where causality, admissibility, reduced observer access, and transport sectors are inherited from the parent framework. Second, there is the route-specific FTL layer, where one asks whether a particular soft readout architecture or hard carrier architecture can outrun the chosen luminal reference sector without violating the admissibility conditions of that framework. Third, there is the pre-engineering layer, where calibration, control, material embedding, and prototype readiness are discussed. The present manuscript belongs primarily to the second layer while making explicit where theorem closure ends and pre-engineering consequences begin.

1.1 Observer-relative meaning of FTL

Within this program, the notation of superluminal transport is always comparative. One first fixes a luminal reference sector L and its effective cone speed c_L in a chosen observer window. One then compares another sector T to L and asks whether the observer-relative transport speed of T exceeds the reference speed determined by L. This is why the glossary below is not merely terminological; it is part of the conceptual discipline of the manuscript. Without a fixed observer window, a fixed reference sector, and an explicit admissibility test, the phrase "faster than light" has no canonical meaning here.

The same principle explains why the manuscript repeatedly returns to coarse-graining, detector access, and observer dependence. Different observers may assign different effective trajectories or even different kinematic descriptions to the same deeper process. Accordingly, the correct target of the FTL program is not an observer-independent metaphysical claim about absolute speed, but a controlled family of observer-relative effective transport statements.

1.2 Soft route and hard route

The program splits immediately into a soft route and a hard route, and the distinction is not cosmetic. Soft FTL concerns remote observer-level readout through a non-factorizable soft link-sector. It aims at witness extraction, scanner structure, repeatable record formation, and eventually hierarchical remote-scene approximation. Hard FTL concerns transport on a carrier branch that outruns the chosen luminal sector while preserving causal admissibility, pole observability, identity constraints, composite binding, and cycle closure. Soft FTL is therefore a readout problem before it is a transport problem, whereas hard FTL is a transport problem whose later calibration and repeatability questions come only after theorem closure.

1.3 What is proved here

The manuscript keeps a strict internal boundary between imported assumptions, proved route-level results, and downstream engineering consequences.

Imported framework content consists of observer-relative semantics, luminal reference sectors, and admissibility conditions. These are fixed before the FTL routes are analyzed.

Proved soft-route content consists of candidate selection, branch-locked sector definition, binary witness structure, explicit and kernel-generated witness families, finite and hierarchical scanner organization, detector-memory operationalization, and the cumulative soft closure package.

Proved hard-route content consists of carrier candidate selection, branch-transfer carrier definition, explicit one-pole transport families, projected-pole survival for the explicit family, identity-preserving carrier cycles, composite binding and composite transport theorems, quantitative robustness margins, and the general hard-closure theorems H-C1 through H-C5 proved from the perturbative lemmas fixed in the hard route.

In particular, the manuscript treats the hard chain H-C1 through H-C5 as proved route-level theorems with explicit positivity, continuity, and projection-compatibility hypotheses, and Chapter 6 records the corresponding dependency-complete proofs.

Downstream integrated and pre-engineering content consists of the minimal soft-hard composition law, mission admissibility, calibration chains, control and resource closure, material embedding, repeatability, and the transition from theorem closure to prototype design.

1.4 Status discipline for manuscript claims

The manuscript uses a strict status discipline.

- Imported framework statements are semantic or admissibility assumptions fixed before the route analysis begins.
- Theorem-like items marked ‘[P]‘ are proved in the manuscript from those imported assumptions and the local route hypotheses stated with them.
- Engineering statements marked ‘[E]‘ are downstream design consequences of theorem closure rather than additional closure theorems.
- Failure modes and non-claims are mandatory parts of the theory statement, not optional caveats added after the fact.

This policy matters because the manuscript documents mix theorem closure, numerical closure, and prototype intent at several points. The clean manuscript must preserve the substantive content

while removing ambiguity about what is proved, what is a downstream engineering consequence, and what remains outside the present claim boundary.

1.5 How this manuscript will use the glossary

The glossary that follows is intentionally narrow. It normalizes only the status labels, the core route vocabulary, and the main theorem ledger needed to read the manuscript as a theory text. It is not a package index and not a numerical archive register. Later chapters rely on these labels without reintroducing them in ad hoc form.

FTL Symbol Glossary

This table normalizes only the vocabulary needed to read the present theory manuscript.

Status Labels

Label	Meaning
‘[I]‘	Imported framework statement fixed before the route analysis.
‘[P]‘	Theorem-level statement proved in the manuscript from its stated hypotheses.
‘[E]‘	Engineering or pre-engineering consequence derived after theorem closure.
Soft FTL	Observer-level remote readout regime without ordinary signal propagation in observer space.
Hard FTL	Transport regime in which a carrier sector outruns the chosen luminal reference sector.
I0	First integrated architecture in which a soft scan configures a hard entry schedule.

Core Theory Vocabulary

Term or symbol	Normalized meaning
L	Reference luminal transport sector defining the effective light-cone speed in a chosen observer window.
T	Transport or carrier sector compared against L in sector-relative FTL claims.
c_L	Effective cone speed of the reference luminal sector.
$v_O(T)$	Observer-relative speed of transport sector T.
$\mathrm{FTL}_O(T; L)$	Predicate asserting that sector T is faster than the chosen luminal reference sector for observer O.
Ω_{AB}	Candidate soft link-sector shared between two observer windows.
Branch-locked sector	Preferred soft-FTL candidate in which local and remote windows read one shared non-factorizable branch structure.
Shared-memory bridge	Weak soft precursor based on persistent distributed memory rather than a genuine shared branch structure.
Carrier cycle	Hard-FTL entry, hold, and exit cycle on a superluminal transport branch.
B_T	Effective carrier-branch quantity used in the hard-transport discussion and benchmarks.
R_{soft}	Soft readout error metric used in the engineering prototype criteria.
$\Delta_{\min}$	Minimum readable margin separating valid soft output classes.

Canonical FTL Statement Ledger

Item	Status	Canonical interpretation
Imported observer-relative semantics	'[I]'	Causal primacy, cone relativity, observer dependence, and admissibility tests govern all soft and hard FTL claims.
Branch-locked witness package	'[P]'	First soft theorem package built around binary distinguishability in a shared branch structure.
Explicit hard carrier chain	'[P]'	Explicit one-pole and composite hard transport theorems proved in the manuscript.
General hard closure chain H-C1–H-C5	'[P]'	Carrier existence, projected-physical observability, identity preservation, composite binding, observability, and identity compatibility, together with robustness lifting, close the hard route as a theorem package under the explicit perturbation, openness, continuity, and projection-compatibility hypotheses fixed in the hard chapter.
Soft/hard split	Core architecture statement	Soft FTL concerns readout; hard FTL concerns transport; they must not be collapsed into one claim.
Integrated architecture I0	'[E]' design target	Soft scan configures hard initialization in a first coupled system.

2 Imported Foundations

This chapter fixes the parent-theory structures that the FTL program is allowed to reuse without alteration. The purpose of this section is not to rederive the full observer framework, but to fix

the imported assumptions used below. In particular, transport speed is not fundamental, observer-accessible kinematics depends on projection and slicing, and a luminal reference sector is required before any superluminal comparison can be stated.

The clean manuscript therefore imports three foundational ingredients. First, it imports the observer layer: an observer is not an abstract spectator but a specific choice of spatial projection, temporal slicing, and decategorified readout regime. Second, it imports the luminal reference layer: a chosen sector L provides the effective cone speed against which another process is compared. Third, it imports the admissibility layer: even if a process is observer-relative superluminal with respect to L, it is not thereby exempt from causal audit, detector readability requirements, or projection-compatibility constraints.

2.1 Causal primacy and effective cones

The first imported principle is causal primacy. The parent framework does not begin with metric speed bounds; it begins with structural order, observer access, and only then derives effective transport language inside a chosen regime. This means that the effective cone attached to a luminal sector is not a metaphysical ceiling on all possible processes. It is an emergent comparison object inside one observer-relative transport description. The imported formal layer records this in the same order: observer first, luminal reference second, effective-velocity comparison third, and only then observer-relative FTL status.

The manuscript uses this distinction to block a recurrent misunderstanding. A statement of the form

$$v_O(T) > c_L$$

is not the claim that the underlying theory has been shown to violate causal structure. It is the claim that, in the observer window O and relative to the chosen luminal reference sector L, the transport sector T outruns the effective cone attached to L. That is a comparative effective statement, not an absolute one.

This imported principle also determines the burden of later chapters. Soft-route arguments must show that remote readout remains causally admissible even though it is operationally faster than ordinary signal propagation in the chosen observer regime. Hard-route arguments must show that carrier cycles outrun the reference luminal sector while retaining transport identity, observability, and admissible control. Both routes therefore presuppose the same causal order but use it differently.

2.2 Observer-relative FTL predicate

The canonical observer-relative FTL predicate has four pieces: an observer, a reference luminal sector, a process code, and a measured effective speed. The imported framework fixes these as the observer object, the reference luminal sector, the observer-relative process, and the observer-relative FTL predicate. Their role in the manuscript is conceptual as much as formal. They force every FTL claim to answer four questions in advance.

1. Which observer window is being used to define the effective readout or transport speed?
2. Which luminal sector is functioning as the comparison standard?
3. Which process is being described: a soft readout process, a hard carrier process, or an integrated mission process?

4. Which admissibility conditions remain in force while that process is evaluated?

Only after these data are fixed does the FTL question become well-posed. This is why the present manuscript refuses the language of anonymous "superluminal channels". A channel is relevant here only after it has been localized to a definite observer-relative comparison problem.

Two immediate consequences follow. First, distinct observers may assign distinct effective speeds to the same deeper process because the observational projections differ. Second, the same process may be superluminal relative to one chosen luminal sector and not superluminal relative to another. The manuscript does not treat these consequences as inconveniences to be hidden. They are part of the exact meaning of FTL in the structural-physical framework and are therefore part of the protocol of every later theorem statement.

With this imported layer fixed, the theory manuscript can proceed cleanly. Section 3 treats soft sectors and remote readout. Section 4 treats hard carriers and transport. Section 5 treats their integration into one mission stack.

These later questions make sense only after the present section fixes the observer-relative semantics of FTL.

3 Soft FTL Route

The soft route is developed in the manuscript as a sequential chain from the first candidate constructions to the soft-closure statement. This chapter now follows that order directly: X.14 candidate classes, X.15 branch-locked definition and binary witness, X.16 explicit two-branch family, X.17 kernel-generated two-branch family, X.18 finite-state automaton, X.19 finite scanner, X.20 operational detector-memory protocol, X.21 hierarchical scanner, and X.22 robustness and closure. Chapter 6 contains the linked prose proofs for the theorem statements that are isolated explicitly below.

3.1 X.14. First Candidate Constructions for a Non-Factorizable Soft Link-Sector

The manuscript begins from the reduced-state layer rather than from exotic geometry. Within the logic of the main theory manuscript, the first candidate for a soft link-sector should live where the formal apparatus is already explicit: admissible subsystem factorization, reduced states, stable detector basis, decoherence, classicalization, and information current. The methodological conclusion is therefore immediate:

the first soft link-sector should be built as a special composite sector in the reduced-state language.

The manuscript then fixes five minimal design criteria. A candidate Ω_{AB} must be observer-accessible, nontrivial at the reduced-state level, readable in a classical detector regime, and non-factorizable in the operational sense. It must also come with an interpretable failure audit. These conditions are the manuscript-level filter used for the first soft program.

The first candidate class is the shared-memory bridge. One writes

$$H = H_A \otimes H_M \otimes H_B \otimes H_E,$$

with M a shared memory sector, A a local detector/readout sector, B a remote coarse sector, and E the environment. A remote coarse variable $b \in \{0, 1\}$ is first written into memory,

$$U_{BM} : |b\rangle_B \otimes |m_*\rangle_M \mapsto |b\rangle_B \otimes |m_b\rangle_M,$$

and then read by the local detector,

$$U_{AM} : |a_*\rangle_A \otimes |m_b\rangle_M \mapsto |r_b\rangle_A \otimes |\widetilde{m}_b\rangle_M.$$

This yields a reduced detector state of the form

$$\rho_A^{(b)} \approx |r_b\rangle\langle r_b|.$$

The manuscript treats this as a soft precursor rather than the first strong theorem package, because the remote dependence has already been written into an intermediate memory sector.

The second candidate class is the branch-locked composite sector. Here one introduces global branch states

$$|\Psi^{(b)}\rangle = \sum_{m=1}^{M} \alpha_m^{(b)} |r_m\rangle_A \otimes |\chi_m^{(b)}\rangle_{BE},$$

so that after tracing over BE one obtains

$$\rho_A^{(b)} = \mathrm{Tr}_{BE}(|\Psi^{(b)}\rangle\langle\Psi^{(b)}|).$$

When branch overlaps are sufficiently small, the effective classical form becomes

$$\rho_A^{(b)} \approx \sum_{m=1}^{M} p_m^{(b)} |r_m\rangle\langle r_m|, \qquad p_m^{(b)} = |\alpha_m^{(b)}|^2.$$

This is the first genuine soft-FTL candidate in the manuscript, because the local detector reads a shared branch structure rather than a prewritten memory bridge.

The third candidate class is the topological soft link-sector. A global topological label $\tau \in \mathcal{T}$ determines the sector, and the local detector reads a response depending on $\tau(b)$. The manuscript treats this as attractive but later, because it depends on many-body and topological layers that are not yet supposed to carry the first soft theorem package.

The fourth candidate class is the chemistry-like information-bearing link. It is framed as a possible bridge toward material realization, but again not as the first object to formalize.

The comparative conclusion of X.14 is the main organizational result of the early soft route.

Theorem 3.1 [P]. Branch-locked composite sectors are the first main-line soft candidate. *Among the normalized soft candidate classes, the branch-locked composite sector is the first main-line candidate because it is the earliest class that remains inside the reduced-state observer layer, supports readable local detector shadows, and does not yet require the later many-body or material layer.*

Its proof is collected in Chapter 6. The theorem says that Candidate II should be pursued first, while Candidate I serves as a control precursor and Candidates III and IV are later strengthening paths.

3.2 X.15. Branch-Locked Soft Sector: Precise Definition and First Binary Witness Theorem

The next source step is no longer organizational. It fixes the precise definition of the branch-locked soft sector. The manuscript builds this object from three previously established ingredients: amplitude states as organized families of coarse-grained amplitudes, observer-level coarse projection onto classes of histories, and reduced detector states in the measurement/decoherence block.

Definition 3.2 Branch-locked soft sector. Fix an observer regime

$$O = (\Pi_O, \Theta_O, \chi_O)$$

and an admissible observer partition of histories into coarse classes. A family of coarse-grained amplitude sectors

$$\mathfrak{B} = \{\mathcal{C}^{(b)}\}_{b\in\Sigma_B}$$

is a branch-locked soft sector if the following hold.

There is a finite remote coarse alphabet Σ_B. For each $b \in \Sigma_B$, there is a normalized amplitude state $\widehat{\Psi}_O^{(b)}$ that admits an observer-access decomposition

$$H = H_A \otimes H_{BE}$$

and takes the form

$$\widehat{\Psi}_O^{(b)} = \sum_{m=1}^{M} \alpha_m^{(b)} |r_m\rangle_A \otimes |\chi_m^{(b)}\rangle_{BE},$$

where $\{|r_m\rangle_A\}_{m=1}^M$ is a fixed local detector basis, the branch states $|\chi_m^{(b)}\rangle_{BE}$ live in the remote-plus-environment sector, and the coefficients depend on the remote coarse label.

The local detector basis is branch-locked: it does not depend on b. The reduced detector shadow

$$\rho_A^{(b)} := \mathrm{Tr}_{BE}(|\widehat{\Psi}_O^{(b)}\rangle\langle\widehat{\Psi}_O^{(b)}|)$$

is not constant in b. There is an effective classicality window in which

$$\rho_A^{(b)} \approx \sum_{m=1}^{M} p_m^{(b)} |r_m\rangle\langle r_m|.$$

Finally, the map $b \mapsto \rho_A^{(b)}$ is not interpreted as ordinary signal flight inside the chosen luminal sector but as the local shadow of one shared branch-locked amplitude structure.

The manuscript then sharpens the meaning of non-factorizability.

Definition 3.3 Soft non-factorizability. A branch-locked soft sector is non-factorizable if there is no admissible re-encoding such that, for all $b \in \Sigma_B$,

$$\widehat{\Psi}_O^{(b)} = |\varphi\rangle_A \otimes |\zeta^{(b)}\rangle_{BE}$$

with one and the same local factor $|\varphi\rangle_A$ independent of b.

In the manuscript, this is an operational definition: if such a factorization existed, the local reduced detector state would be fixed and no soft witness would be possible.

The first numerical invariant is the branch-weight vector

$$p^{(b)} = (p_1^{(b)}, \ldots, p_M^{(b)}), \qquad p_m^{(b)} = \langle r_m|\rho_A^{(b)}|r_m\rangle,$$

and for a pair of remote labels (b_0, b_1) the manuscript defines the binary witness contrast by

$$\Delta_{b_0,b_1} := \frac{1}{2}\|\rho_A^{(b_0)} - \rho_A^{(b_1)}\|_1.$$

In the diagonal detector regime, this becomes

$$\Delta_{b_0,b_1} = \frac{1}{2}\sum_{m=1}^{M} |p_m^{(b_0)} - p_m^{(b_1)}|.$$

The manuscript then extracts the first derived proposition: in the effective classicality window the reduced detector state is approximately diagonal in the stable pointer basis. That is the minimal condition for ordinary local readout.

The first theorem package of X.15 is then the binary witness theorem.

Theorem 3.4 [P]. First binary witness theorem. *Let*

$$\mathfrak{B} = \{\mathcal{C}^{(b)}\}_{b\in\Sigma_B}$$

be a branch-locked soft sector. Assume that there exists a pair $b_0 \neq b_1$ with

$$\Delta_{b_0,b_1} > 0,$$

that the detector basis is a stable pointer basis, and that the classicalization error is sufficiently small in some readout window. Then there exists a local detector observable

$$M_A = \sum_{m=1}^{M} \mu_m |r_m\rangle\langle r_m|$$

and a readout time t_* such that

$$\mathrm{Tr}(M_A \rho_A^{(b_0)}(t_*)) \neq \mathrm{Tr}(M_A \rho_A^{(b_1)}(t_*)).$$

In other words, the branch-locked sector realizes a binary soft witness protocol.

The proof is in Chapter 6. The theorem is route-level and exact relative to the stated branch-locked sector hypotheses.

X.15 also records the branch-locked failure modes explicitly: no local shadow at all, no stable common pointer basis, failure of classicalization, admissible refactorization that kills the soft effect, and readout differences that exist formally but remain below operational threshold after noise.

3.3 X.16. Explicit Two-Branch Family and the First Concrete Binary Soft Witness Model

The next source step is to stop speaking only abstractly and exhibit the first fully explicit two-branch family. Fix an observer-level factorization

$$H = H_A \otimes H_{BE},$$

and a stable local detector basis

$$\{|r_1\rangle, |r_2\rangle\}.$$

The explicit family is

$$\widehat{\Psi}_O^{(0)} = \sqrt{p}|r_1\rangle \otimes |\chi_1\rangle + \sqrt{1-p}|r_2\rangle \otimes |\chi_2\rangle,$$
$$\widehat{\Psi}_O^{(1)} = \sqrt{q}|r_1\rangle \otimes |\eta_1\rangle + \sqrt{1-q}|r_2\rangle \otimes |\eta_2\rangle,$$

with $0 \leq p, q \leq 1$ and normalized branch states in H_{BE}.

Tracing over BE gives

$$\rho_A^{(0)} = \begin{pmatrix} p & \sqrt{p(1-p)}\,\gamma_0 \\ \sqrt{p(1-p)}\,\overline{\gamma_0} & 1-p \end{pmatrix}, \qquad \gamma_0 := \langle \chi_2, \chi_1\rangle,$$

and

$$\rho_A^{(1)} = \begin{pmatrix} q & \sqrt{q(1-q)}\,\gamma_1 \\ \sqrt{q(1-q)}\,\overline{\gamma_1} & 1-q \end{pmatrix}, \qquad \gamma_1 := \langle \eta_2, \eta_1 \rangle,$$

with the overlap constraints

$$|\gamma_0| \le 1, \qquad |\gamma_1| \le 1.$$

This explicit model already shows that the remote label can influence the local detector shadow through two channels: diagonal branch weights and off-diagonal coherence data.

In the diagonal detector limit one has

$$\rho_A^{(0)} \approx \begin{pmatrix} p & 0 \\ 0 & 1-p \end{pmatrix}, \qquad \rho_A^{(1)} \approx \begin{pmatrix} q & 0 \\ 0 & 1-q \end{pmatrix},$$

so the binary witness contrast becomes

$$\Delta_{0,1} = |p - q|.$$

In the full two-by-two case, the manuscript computes

$$\Delta_{0,1} = \sqrt{(p-q)^2 + \left| \sqrt{p(1-p)}\,\gamma_0 - \sqrt{q(1-q)}\,\gamma_1 \right|^2}.$$

This formula is exact for the Hermitian two-by-two reduced detector states displayed above.

This yields the first concrete binary witness criterion: the witness exists if and only if the pair of reduced detector states is distinct, equivalently if and only if the displayed formula is strictly positive.

The manuscript then distinguishes three regimes. In the purely classical branch-weight regime, $\gamma_0 = \gamma_1 = 0$ and the witness is exactly $|p - q|$. In the purely coherence-based regime, one can have $p = q$ but a nonzero difference in the coherence term, so the witness survives before full classicalization. In the mixed regime, both contributions are present.

The explicit family also gives the first direct non-factorizability criterion.

Proposition 3.5 [P]. Explicit non-factorizability criterion. *If the two-branch family satisfies*

$$\Delta_{0,1} > 0,$$

then it cannot be brought by an admissible re-encoding to the form

$$\widehat{\Psi}_O^{(b)} = |\varphi\rangle_A \otimes |\zeta^{(b)}\rangle_{BE}$$

with a common detector factor $|\varphi\rangle_A$ independent of b.

X.16 is the first explicit calculable theorem block for Candidate II. What X.17 adds is not theorem status, but derivation of the same structure from the kernel/resolvent language itself.

3.4 X.17. Kernel-Generated Two-Branch Soft Family and the First Robustness Question

X.17 is the first source step that lifts the two-branch family from an ansatz to a kernel-generated object. Fix the observer's amplitude-marked kernel

$$K_O^{\mathrm{amp}}(z, \theta, \beta, \phi) = \sum_e \alpha(e) z^{\gamma(e)} e^{i\theta \cdot \Delta x_O(e)} e^{-\beta a(e)} e^{i\phi \xi_D(e)} [e],$$

and the full resolvent amplitude

$$\Psi_O(z,\theta,\beta,\phi) = \iota^\top (I - K_O^{\mathrm{amp}}(z,\theta,\beta,\phi))^{-1}\omega.$$

Choose two coarse classes C_1 and C_2 and define, for each remote label $b \in \{0,1\}$,

$$A_1^{(b)} := \Psi_O^{(b)}(C_1), \qquad A_2^{(b)} := \Psi_O^{(b)}(C_2),$$

where the superscript b indicates evaluation at different remote-dependent kernel marks, for example through $\phi = \phi_b$.

The resulting two-component amplitude state is

$$\Psi_O^{(b)} = (A_1^{(b)}, A_2^{(b)}),$$

and its normalized version is

$$\widehat{\Psi}_O^{(b)} = \frac{A_1^{(b)}|r_1\rangle_A \otimes |\chi_1^{(b)}\rangle_{BE} + A_2^{(b)}|r_2\rangle_A \otimes |\chi_2^{(b)}\rangle_{BE}}{\sqrt{|A_1^{(b)}|^2 + |A_2^{(b)}|^2}}.$$

With

$$N_b := |A_1^{(b)}|^2 + |A_2^{(b)}|^2, \qquad p_b := \frac{|A_1^{(b)}|^2}{N_b},$$

the reduced detector state again takes the form

$$\rho_A^{(b)} = \begin{pmatrix} p_b & \sqrt{p_b(1-p_b)}\,\gamma_b \\ \sqrt{p_b(1-p_b)}\,\overline{\gamma_b} & 1-p_b \end{pmatrix},$$

where

$$\gamma_b = e^{i(\phi_1^{(b)} - \phi_2^{(b)})}\langle \chi_2^{(b)}, \chi_1^{(b)}\rangle, \qquad |\gamma_b| \le 1.$$

Here $\phi_1^{(b)}$ and $\phi_2^{(b)}$ denote the phases of the coarse amplitudes $A_1^{(b)}$ and $A_2^{(b)}$, respectively.

The first kernel-generated binary witness criterion is then simply the X.16 witness formula with the parameters expressed through coarse-grained kernel amplitudes. In the manuscript, this establishes the first true bridge from amplitude-marked kernels and coarse projection to a soft binary witness.

The section also distinguishes phase-soft, weight-soft, and mixed-soft sectors. In the pure phase-soft case the kernel changes phases but not magnitudes. In the weight-soft case the magnitudes themselves change, so the witness survives more naturally in the diagonal detector limit. This is the point where the manuscript begins to privilege weight-soft sectors for the first robust theorem chain.

The last object of X.17 is the first robustness question. If the kernel is perturbed by an admissible small deformation

$$K_O^{\mathrm{amp}} \mapsto K_O^{\mathrm{amp}} + \delta K,$$

and the coarse partition and detector basis remain fixed, does the inequality

$$\Delta_{0,1} > 0$$

survive? The manuscript does not claim a global theorem yet. It only isolates a local openness statement: if the initial witness margin is positive and no branch collision collapses both the diagonal and coherence contributions simultaneously, then sufficiently small admissible perturbations preserve the witness.

3.5 X.18. Finite-State Observer-Amplitude Automaton for the First Soft Witness Sector

The next source move is to replace the generic two-class construction by the smallest finite-state automaton in which the soft witness is exact and fully calculable. The automaton is

$$\mathcal{A}^{(2)}_{\mathrm{soft}}, \qquad B = \{I, T\}, \qquad \iota = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \omega = \begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

with exactly two admissible elementary branches

$$e_1 : I \to T, \qquad e_2 : I \to T, \qquad \xi_D(e_1) = 1, \qquad \xi_D(e_2) = 2.$$

For each remote coarse label $b \in \{0, 1\}$ one assigns branch coefficients $a_1^{(b)}$ and $a_2^{(b)}$, giving the kernel

$$K^{(b)} = \begin{pmatrix} 0 & a_1^{(b)}[e_1] + a_2^{(b)}[e_2] \\ 0 & 0 \end{pmatrix}.$$

Because the automaton has no loops,

$$(K^{(b)})^2 = 0, \qquad (I - K^{(b)})^{-1} = I + K^{(b)}.$$

Thus the full amplitude object is

$$\iota^{\top}(I - K^{(b)})^{-1}\omega = a_1^{(b)}[e_1] + a_2^{(b)}[e_2].$$

Defining the coarse classes $C_1 = \{\Gamma : e_1\}$ and $C_2 = \{\Gamma : e_2\}$ gives

$$A_1^{(b)} = a_1^{(b)}, \qquad A_2^{(b)} = a_2^{(b)},$$

so that the normalized amplitude state and reduced detector state are exactly the two-branch objects already used above. This is the manuscript's first completely explicit finite-state binary witness sector.

The first weight-soft specialization is

$$a_1^{(0)} = a(1 + \varepsilon), \qquad a_2^{(0)} = a(1 - \varepsilon),$$

$$a_1^{(1)} = a(1 - \varepsilon), \qquad a_2^{(1)} = a(1 + \varepsilon),$$

with $a > 0$ and $0 < \varepsilon < 1$. In the diagonal detector limit,

$$p_0 = \frac{(1 + \varepsilon)^2}{2(1 + \varepsilon^2)}, \qquad p_1 = \frac{(1 - \varepsilon)^2}{2(1 + \varepsilon^2)},$$

and therefore

$$\Delta_{0,1} = |p_0 - p_1| = \frac{2\varepsilon}{1 + \varepsilon^2} > 0.$$

This is the first concrete binary witness theorem of the finite-state soft route.

Theorem 3.6 [P]. Concrete binary witness theorem for the minimal automaton. *Let $\mathcal{A}^{(2)}_{\mathrm{soft}}$ be the minimal finite-state observer-amplitude automaton with the weight-soft deformation displayed above. If the detector is in the diagonal detector limit, then the automaton realizes a binary soft witness sector with*

$$\Delta_{0,1} = \frac{2\varepsilon}{1 + \varepsilon^2} > 0.$$

The proof is collected in Chapter 6. The manuscript immediately compares this to the pure phase-soft specialization, where only a relative phase changes and the diagonal witness disappears after full classicalization. That is why weight-soft automata become the main line for the first operational theorem package.

X.18 also records the first local robustness criterion. If detector weights drift by δ_0 and δ_1, then the diagonal witness survives whenever

$$|\delta_0| + |\delta_1| < \frac{2\varepsilon}{1+\varepsilon^2}.$$

In the symmetric case $|\delta_0|, |\delta_1| \le \delta$, it suffices that

$$\delta < \frac{\varepsilon}{1+\varepsilon^2}.$$

The manuscript uses this as the first quantitative checkpoint of soft robustness.

3.6 X.19. Finite-Alphabet Extension and the First Scanner Theorem

The next source section lifts the binary witness automaton to a finite scanner. The automaton remains minimal in state set,

$$B = \{I, T\},$$

but there are now N admissible branches

$$e_1, \dots, e_N : I \to T, \qquad \xi_D(e_k) = k.$$

For each remote label $b \in \Sigma_B$ one assigns amplitudes $a_k^{(b)}$, giving

$$K^{(b)} = \begin{pmatrix} 0 & \sum_{k=1}^{N} a_k^{(b)}[e_k] \\ 0 & 0 \end{pmatrix}, \qquad (K^{(b)})^2 = 0, \qquad (I - K^{(b)})^{-1} = I + K^{(b)}.$$

The coarse amplitudes are

$$A_k^{(b)} = a_k^{(b)}, \qquad N_b = \sum_{j=1}^{N} |a_j^{(b)}|^2, \qquad p_k^{(b)} = \frac{|a_k^{(b)}|^2}{N_b}.$$

The first finite scanner theorem is again weight-soft. For $\Sigma_B = \{1, \dots, N\}$,

$$a_k^{(b)} = \begin{cases} a(1+\varepsilon), & k = b, \\ a(1-\varepsilon), & k \neq b, \end{cases} \qquad a > 0, \quad 0 < \varepsilon < 1.$$

Then

$$D_N(\varepsilon) = (1+\varepsilon)^2 + (N-1)(1-\varepsilon)^2,$$

and in the diagonal detector limit one has

$$p_k^{(b)} = \begin{cases} \dfrac{(1+\varepsilon)^2}{D_N(\varepsilon)}, & k = b, \\ \dfrac{(1-\varepsilon)^2}{D_N(\varepsilon)}, & k \neq b. \end{cases}$$

For $b \neq c$ the pairwise scanner contrast is

$$\Delta_{b,c}^{(N)} = \frac{4\varepsilon}{(1+\varepsilon)^2 + (N-1)(1-\varepsilon)^2} > 0.$$

This is the manuscript's first finite scanner theorem. It is accompanied immediately by the scaling law

$$\Delta_{b,c}^{(N)} \sim \frac{4\varepsilon}{N(1-\varepsilon)^2} \qquad \text{as } N \to \infty,$$

which the manuscript uses to block naive extrapolation from finite scanner to arbitrarily high-resolution remote image.

The first robustness margin of the finite scanner is also explicit. If detector probabilities are perturbed by $\delta_k^{(b)}$ with

$$\sum_{k=1}^{N} |\delta_k^{(b)}| \leq \eta,$$

then injectivity survives whenever

$$\eta < \Delta_{\min}^{(N)} = \frac{4\varepsilon}{(1+\varepsilon)^2 + (N-1)(1-\varepsilon)^2}.$$

3.7 X.20. Scanner Readout Dynamics, Memory Persistence, and the First Operational Soft Protocol

After X.19 the manuscript insists that the theorem chain is still incomplete. One has a finite scanner, but only statically. The next step is to pass through reduced dynamical maps, detector classicalization, observer memory, information gain, and selective readout.

Fix an observer-accessible decomposition

$$H = H_S \otimes H_D \otimes H_E,$$

with S the scanned remote-linked sector, D the detector sector, and E the environment. Inside H_D choose an observer memory sector

$$H_{\text{mem}} \subset H_D,$$

whose stable basis states store persistent records. For a finite remote alphabet $\Sigma_B = \{1, \ldots, N\}$, the manuscript introduces a family of reduced detector maps

$$\Phi_t^{(b)} : \rho_D^{\text{blank}} \mapsto \rho_D^{(b)}(t),$$

and requires detector classicalization,

$$\epsilon_{\text{cl}}^{(b)}(t) = \|\rho_D^{(b)}(t) - \rho_D^{(b),\text{cl}}(t)\|_1 \ll 1,$$

followed by memory write into states $|m_b\rangle$.

The manuscript defines the first operational finite soft scanner protocol as the sequence: prepare blank detector and memory, couple through the reduced scanner map, wait for classicalization, write to memory, and read out selectively. The key point is that the scanner becomes a persistent local record rather than only a transient reduced shadow.

The first operational theorem then says: if the finite soft sector exists, the pointer basis is stable, classicalization reaches a sufficiently small error in some readout window, and memory states are pairwise distinguishable, then there is an operational selective soft scanner protocol on Σ_B.

The first quantitative operational inequality is

$$\epsilon_{\max}(t_*) + \eta_{\mathrm{mem}} < \Delta_{\min}^{(N)}.$$

The manuscript immediately uses this to identify the first operational failure modes: detector never becomes classical enough, memory write fails, selective information is washed out by unselective averaging, memory persistence is too short, or reset/erasure costs destroy repeatable use.

Proposition 3.7 [P]. Soft background yields the minimal soft theorem package. *If the soft witness background holds, then the program has both a branch-locked binary witness and a repeatable soft witness regime. Equivalently, the minimal soft background recovers both branch-locked binary distinguishability and soft repeatability.*

The proof is in Chapter 6. In source terms, this is the first point where the soft branch is no longer only a collection of constructions but a route-level theorem package.

3.8 X.21. Hierarchical Soft Scanner, Multi-Scale Labels, and the First Remote-Scene Approximation

The manuscript next argues that a finite flat alphabet is not enough if the soft route is supposed to approach remote-scene readout. One must pass to a hierarchy of coarse labels. A hierarchical remote label is

$$\mathbf{b} = (b^{(1)}, b^{(2)}, \ldots, b^{(L)}),$$

where $b^{(\ell)}$ is the coarse description at scale ℓ, and finer labels refine coarser ones.

For each scale $\ell = 1, \ldots, L$, the manuscript introduces its own finite alphabet

$$\Sigma^{(\ell)} = \{1, \ldots, N_\ell\}$$

and its own soft scanner automaton. The full hierarchical scanner is the family

$$\mathfrak{S}_{\mathrm{hier}} = \{\mathcal{A}_{\mathrm{soft}}^{(1)}, \ldots, \mathcal{A}_{\mathrm{soft}}^{(L)}\}.$$

In the simplest weight-soft model, each scale carries its own parameters a_ℓ and ε_ℓ, so that the scale-wise minimal contrast is

$$\Delta_{\min}^{(\ell)} = \frac{4\varepsilon_\ell}{(1+\varepsilon_\ell)^2 + (N_\ell - 1)(1-\varepsilon_\ell)^2}.$$

If scale-wise detector states factorize at leading order, a natural hierarchical contrast is the weighted sum of scale-wise contrasts. The first hierarchical scanner theorem then states that if each scale is readable in the diagonal detector limit, the map

$$\mathbf{b} \mapsto \rho_{D,\mathrm{hier}}^{(\mathbf{b})}$$

is injective on the set of consistent multi-scale labels.

The manuscript interprets this as the first remote-scene approximation: not unrestricted image formation, but a coherent multi-scale coarse description of the remote sector. Memory persistence is then lifted scale by scale by decomposing the observer memory sector into submemories, one for each scale.

The first operational remote-scene theorem says that if each scale admits a finite weight-soft scanner, classicalization succeeds on all levels, memory write succeeds on all levels, and scale-wise memory records are distinguishable, then there is an operational hierarchical soft scanner protocol reconstructing the full multi-scale label **b**.

The scale-wise quantitative viability criterion is

$$\epsilon_{\max}^{(\ell)}(t_*) + \eta_{\mathrm{mem}}^{(\ell)} < \Delta_{\min}^{(\ell)} \qquad \text{for all } \ell = 1, \dots, L.$$

The manuscript uses this to show that remote-scene approximation is naturally hierarchical rather than flatly high-resolution.

3.9 X.22. Robustness of the Hierarchical Soft Scanner and the Soft-Closure Statement

The final source step on the soft side is the robustness theorem and closure statement. The manuscript explicitly asks for an audit-level formulation: the soft branch should not depend on one fine-tuned microscopic presentation but should survive admissible re-encoding, small detector/readout perturbations, and small deformations of the branch amplitudes.

An admissible soft perturbation at scale ℓ is decomposed into four pieces,

$$\delta K^{(\ell)}, \qquad \delta C^{(\ell)}, \qquad \delta D^{(\ell)}, \qquad \delta M^{(\ell)},$$

corresponding respectively to perturbations of the amplitude-marked kernel, the coarse partition, the detector classicalization dynamics, and the memory-write channel. The perturbation is admissible only if the observer regime stays fixed, the coarse partition does not jump in combinatorial type, the pointer basis remains stable, and the memory sector remains a stable record subspace.

The scale-wise total robustness load is then defined by

$$R_\ell := \zeta_{\mathrm{str}}^{(\ell)} + \epsilon_{\max}^{(\ell)}(t_*) + \eta_{\mathrm{mem}}^{(\ell)},$$

where $\zeta_{\mathrm{str}}^{(\ell)}$ is the purely structural perturbation budget. The local robustness proposition says that if

$$R_\ell < \Delta_{\min}^{(\ell)},$$

then the scanner remains injective on the alphabet at scale ℓ.

The hierarchical robustness theorem is the scale-wise lifting of that statement: if the inequality above holds for every scale, then each scale remains readable, the full multi-scale label remains reconstructible, and the hierarchical scanner continues to provide a stable remote-scene approximation.

This is the manuscript's soft-closure point. The branch is considered closed, at level L, only if the following five ingredients all exist together: an admissible branch-locked scanner sector, a binary witness theorem, a finite and then hierarchical scanner theorem, operational detector-memory readout, and robustness with positive scale-wise margins.

The resulting closure statement is the manuscript's final claim on the soft side:

soft FTL is closed as a robust theorem package for hierarchical remote-scene approximation.

The manuscript is equally explicit about what this does not mean. It is not yet a theorem of unrestricted full-resolution remote vision, nor a theorem of arbitrary detector-basis robustness, nor a material embedding theorem for every possible realization. It is a closed theorem package for hierarchical remote-scene approximation.

The route-level cumulative structure is recorded explicitly in the public package.

Proposition 3.8 [P]. The scanner ladder is cumulative. *Each later soft milestone preserves the witness content of the earlier ones while adding new organizational structure. In particular, finite scanners extend binary witness architectures, and hierarchical scanners extend finite scanners by adding readable multi-scale organization rather than replacing the earlier witness conditions.*

The proof is in Chapter 6. This proposition is the clean source-level summary of the order

$$S0 \to S1 \to S2 \to S3 \to S4,$$

from binary witness through repeatable scanner and finite scanner to hierarchical remote-scene approximation with robustness.

4 Hard FTL Route

This chapter is now written in the same source-order style as the soft route. Rather than compressing the hard program into a short synthetic essay, we follow the sequence of the manuscript text directly: first the candidate carrier classes, then the branch-transfer family, then the explicit one-pole model, then projected-resolvent observability, identity preservation, composite binding, explicit composite transport, robustness, and finally the general hard-closure chain culminating in audited open-class stability.

The manuscript makes the central difference from the soft branch explicit from the outset. Soft FTL can close at the level of observer-accessible remote-scene approximation. Hard FTL asks a different question: whether an object-sector can enter, traverse, and leave a carrier branch whose effective transport speed exceeds the chosen luminal reference while remaining a physically admissible transported object rather than dissolving into an internal or destructive excitation. That is why the hard route must be built as a theorem chain rather than as a slogan about "moving faster than light." In the present chapter this chain is closed by combining explicit carrier constructions with quantitative perturbative lemmas and the general hard-closure theorems H-C1 through H-C5.

4.1 X.23. First Candidate Constructions for a Superluminal Hard Carrier Sector

4.1.1 X.23.1. Why the hard branch must begin with carrier candidates

After X.22, the soft branch is already closed as a robust theorem package for hierarchical remote-scene approximation. Hard FTL must therefore be stated with equal precision. But in the logic of the manuscript it cannot begin with the sentence "move a physical body faster than light." One must first isolate candidate carrier sectors, because the relativistic and field-theoretic layers are not inserted as axioms of spacetime. They arise from transport sectors, kernel-resolvent data, low-momentum spectral branches, and the emergence of an effective cone speed.

So the hard branch begins with the question

$$\text{which transport sectors can even play the role of a carrier } T \text{ with } v_T > c_L?$$

Only after that can one ask whether such a sector can carry a stable object-sector, survive branch transfer, preserve pole and identity, and hold a bound object together.

4.1.2 X.23.2. Minimal design criteria for a hard carrier

The manuscript fixes five requirements immediately.

First, the candidate must be a transport sector in the strict sense of the theory: a stable class of excitations selected by a marked or reduced kernel block and governed by a dominant spectral branch or an equivalent dispersion relation.

Second, it must admit an effective superluminal window,

$$v_T(q) > c_L,$$

on a nonempty momentum range relative to the chosen luminal sector.

Third, it must be object-carrying rather than merely fast. Later field theory reads physical one-particle sectors through pole-residue data, so a fast sector is useless if no physical pole survives in it.

Fourth, it must admit controlled branch accessibility from the ordinary sector; otherwise hard transport cannot even be posed as a cycle problem.

Fifth, for composite objects it must not destroy binding. Boundness and negative binding energy are separate stability facts in the underlying framework, not automatic properties of any fast branch.

Definition 4.1 Minimal hard carrier design. A hard carrier design is admissible only if it supplies a transport sector, a nonempty superluminal window relative to the chosen luminal sector, object-carrying accessibility, controlled branch transfer, and compatibility with later observability and binding constraints.

4.1.3 X.23.3. Candidate class I: the desynchronized transport sector

The earliest and most natural candidate is a desynchronized transport sector. The relativistic transport layer says that ordinary luminal behavior arises when several transport sectors synchronize into quadratic dispersion laws with a common effective cone speed. This suggests the first hard candidate: a sector that does not synchronize into the same cone as the luminal family. In schematic form,

$$P_T(\omega, q) = A_T\omega^2 - B_T|q|^2 - M_T^2 + \cdots, \qquad c_T^2 := \frac{B_T}{A_T}, \qquad c_T > c_L.$$

This candidate is natural because it requires almost no new language beyond transport sectors, effective dispersion laws, and emergent cone speeds. Its weakness is also immediate: it is a fast sector, but not yet a transport protocol for a stable object.

4.1.4 X.23.4. Candidate class II: the branch-transfer carrier from an avoided crossing

The second candidate is more physical and more promising. One takes a luminal kernel block and a potentially superluminal kernel block, then connects them by a switching block so that the effective inverse kernel becomes

$$\mathcal{D}(\omega, q) = \begin{pmatrix} \omega - E_L(q) & -\Delta(q) \\ -\Delta(q) & \omega - E_T(q) \end{pmatrix}.$$

This is already stronger than class I because it contains not only a fast branch but a built-in entry-exit pattern

$$\mathfrak{p}_L \to \mathfrak{p}_T \to \mathfrak{p}_L.$$

It is therefore the first candidate that can lead directly to a one-pole hard theorem package.

4.1.5 X.23.5. Candidate class III: the paired or many-body carrier

The third candidate is a paired, collective, or many-body carrier. Later many-body, paired-phase, and topological sectors suggest that a bound object may ultimately be transported more naturally through a collective regime than through a bare one-pole branch. But this candidate depends on later theorem debt. It should not be the first hard closure target.

4.1.6 X.23.6. Candidate class IV: the projection-shortcut pseudo-carrier

The fourth candidate is not a main-line mechanism but a negative control. Because the theory separates projected geometry from deeper rewriting dynamics, one can imagine a regime where observer-level geometry gives a very short path between two regions, so that transport only looks superluminal. But if such a regime yields no stable carrier branch with pole survival, then it must be classified as an observer-level FTL image rather than as hard transport.

4.1.7 X.23.7. Comparison of the candidate classes

Candidate class I is the earliest and most natural but yields only a fast branch. Candidate class II is the first serious hard candidate because it already contains a branch-transfer structure. Candidate class III is attractive for composite transport but belongs to a later stage. Candidate class IV is required as a control so that projection effects are not mistaken for real carrier transport.

The main conclusion of the section is therefore

the next real hard step is to build candidate II, using candidate I as background and candidate IV as control.

Theorem 4.2 [P]. The branch-transfer carrier is the first main-line hard candidate. *Among the normalized hard carrier classes, the branch-transfer carrier is the first main-line hard candidate because it is the earliest class that combines an explicit superluminal transport window with controlled branch accessibility and direct compatibility with one-pole identity criteria.*

The manuscript is strict about this ordering because a desynchronized sector can be fast while failing every object-preservation test, a many-body carrier can be physically suggestive while importing too much later machinery, and a projection shortcut can mimic transport without supplying a real carrier pole. Candidate II is the first point at which the hard route becomes a theorem program rather than a catalogue of possibilities.

4.1.8 X.23.8. Theorem packages attached to the candidates

The candidate survey already suggests the packages to be built.

Package I is a desynchronized-sector existence theorem whose goal is to show that a transport sector exists with cone speed strictly above the chosen luminal sector.

Package II is the branch-transfer one-pole theorem: existence of a superluminal branch, controlled transfer, pole survival, and small identity loss.

Package III is the composite carrier theorem: a many-body or paired carrier regime that sustains bound objects with positive binding margin.

Proposition 4.3 [P]. Hard background yields the minimal hard theorem package. *If the hard-cycle background holds, then the program has both one-pole hard-cycle admissibility and composite hard-cycle repeatability. In particular, the bundled hard background recovers the minimal one-pole and composite hard theorem package.*

This proposition states the structural significance of the candidate comparison: the hard route is already normalized as a joint one-pole-plus-composite program, not as disconnected subproblems.

4.2 X.24. Branch-Transfer Hard Carrier: Precise Definition and the First One-Pole Transport Problem

4.2.1 X.24.1. Why the hard carrier must be defined through transport sectors and poles

The manuscript now passes from candidate classes to a precise carrier definition. Hard FTL cannot be defined as a fast path in geometry. The relativistic and field-theoretic layers insist that transport sectors are first selected through kernel-resolvent data and only then interpreted through pole-residue language. So a hard carrier is not merely a fast branch. It is a superluminal transport sector that supports a stable one-pole branch and admits controlled branch transfer.

4.2.2 X.24.2. Background data for the hard problem

Fix an observer regime

$$O = (\Pi_O, \Theta_O, \chi_O)$$

and a reference luminal sector L whose low-momentum branch defines the effective cone speed c_L.

An object-sector $\mathfrak{p}$ is a stable one-pole sector if its projected effective two-point object has an isolated simple pole with nonzero residue. A branch is physically admissible only if the projected physical resolvent contains an isolated simple pole with nonzero residue. Internal branch existence alone is not enough.

4.2.3 X.24.3. Precise definition of a branch-transfer hard carrier

One now takes a one-parameter family of effective inverse kernels

$$\mathcal{D}_\lambda(\omega, q), \qquad \lambda \in [0,1],$$

with $\lambda = 0$ the ordinary luminal branch and $\lambda = 1$ the candidate carrier regime.

Definition 4.4 Branch-transfer hard carrier family. The family $\{\mathcal{D}_\lambda\}_{\lambda\in[0,1]}$ is a branch-transfer hard carrier family if:

1. at $\lambda = 0$ it admits an object-carrying branch $\omega = E_L(q)$ with reference cone speed c_L;
2. at $\lambda = 1$ it admits a candidate carrier branch $\omega = E_T(q)$ such that $|\partial_q E_T(q)| > c_L$ on a nonempty window;
3. there exists a continuous selected branch $E_*(q, \lambda)$ connecting E_L to E_T;
4. along the path, the projected physical resolvent keeps an isolated simple pole with nonzero residue;
5. the full cycle $\mathfrak{p}_L \to \mathfrak{p}_T \to \mathfrak{p}_L$ has small identity loss;
6. the switching protocol can be chosen so that transport follows the selected branch rather than leaking uncontrollably into competing branches.

This definition is the first hard analogue of the branch-locked soft-sector definition. It explicitly sews together transport-sector language and pole-residue language.

4.2.4 X.24.4. The avoided-crossing model as the canonical local form

The manuscript then introduces the canonical local form:

$$\mathcal{D}_\lambda(\omega, q) = \begin{pmatrix} \omega - E_L(q) & -\lambda\,\Delta(q) \\ -\lambda\,\overline{\Delta(q)} & \omega - E_T(q) \end{pmatrix}, \qquad \lambda \in [0,1].$$

Its eigenbranches are

$$E_\pm(q,\lambda) = \frac{E_L(q)+E_T(q)}{2} \pm \sqrt{\left(\frac{E_L(q)-E_T(q)}{2}\right)^2 + \lambda^2|\Delta(q)|^2}.$$

This is the canonical branch-transfer normal form.

4.2.5 X.24.5. The first one-pole transport problem

The first hard one-pole transport problem can now be posed exactly. Given a stable one-pole sector on the luminal branch, a candidate carrier branch, and a switching family of the avoided-crossing type, determine whether there exist a nonempty transport window U_T, a switching protocol $\lambda(t)$, and a selected branch $E_*(q,\lambda)$ such that

$$|\partial_q E_*(q,1)| > c_L \quad \text{for } q \in U_T,$$

the object remains a physical one-pole sector along the path, and the cycle identity loss remains small.

4.2.6 X.24.6. First superluminality criterion

The carrier superluminal window is

$$U_T := \{q : \ |\partial_q E_*(q,1)| > c_L\}.$$

All derivative-based statements in the hard route are understood on

$$U_T \cap W_{\mathrm{tr}} \subseteq \{q : q \geq q_{\min} > 0\},$$

so that the carrier derivatives are read away from the surrogate cusp. If for some q_* one already has $|\partial_q E_T(q_*)| > c_L$ and the mixing is sufficiently small relative to the branch separation away from the local crossing, then one of the mixed branches still satisfies the same inequality on a reduced but nonempty neighborhood. This is only a local spectral proposition, but it is the first explicit superluminality criterion.

4.2.7 X.24.7. Pole-survival criterion

Pole survival along transfer means that for each $\lambda \in [0,1]$ the projected physical resolvent of the transported sector has an isolated simple pole

$$\omega = E_*(q,\lambda)$$

with residue

$$Z_*(q,\lambda) \neq 0.$$

This pole-survival criterion is the first closed step in the hard theorem chain: it isolates the exact condition later used in the audited open-class lifting theorem.

4.2.8 X.24.8. Identity preservation as a separate theorem step

The manuscript separates identity preservation from pole survival. One introduces a structural hard identity metric

$$\epsilon_h^{\mathrm{cycle}} = w_\rho d_\rho + w_Q d_Q + w_{\mathrm{struct}} d_{\mathrm{struct}},$$

so that a cycle may preserve a pole while still returning the wrong object. Hard FTL requires both pole survival and identity preservation.

4.2.9 X.24.9. Adiabatic branch accessibility

If $\lambda = \lambda(t)$, define the adiabatic switching score

$$\mathcal{A}(q,t) = \frac{|\dot{\lambda}(t)\,\Delta(q)|}{(E_+(q,\lambda) - E_-(q,\lambda))^2}, \qquad \mathcal{A}_{\max} = \sup_{q,t} \mathcal{A}(q,t).$$

If $\mathcal{A}_{\max} \ll 1$, the manuscript reads branch transfer as adiabatic in the usual observer-level spectral sense: leakage into the unwanted branch is small.

4.2.10 X.24.10. The first one-pole hard theorem [P]

The first one-pole hard theorem can now be stated: if the branch-transfer family has a nonempty superluminal window, pole survival along the selected branch, adiabatic accessibility, and small cycle identity loss, then it realizes a one-pole hard transport protocol

$$\mathfrak{p}_L \to \mathfrak{p}_T \to \mathfrak{p}_L$$

with effective carrier speed above the luminal reference inside the carrier window. Later closure steps generalize this explicit theorem from the local branch-transfer family to the audited open hard class.

4.2.11 X.24.11. The negative control: projection shortcut is not enough

If apparent superluminal displacement is explained only by projection-shortcut geometry and not by a physical one-pole carrier branch in the projected physical resolvent, then it does not qualify as hard transport. This negative control is required so that geometry alone is not mistaken for a carrier theorem.

4.3 X.25. Explicit Two-Branch Hard Carrier Family and the First One-Pole Transport Model

4.3.1 X.25.1. Why the first hard model must be two-branch and explicit

After the precise carrier definition, the manuscript takes the smallest explicit family that can actually be computed. The first hard model must have one luminal branch, one candidate carrier branch, and one switching block. That is the hard analogue of the first explicit soft witness models.

4.3.2 X.25.2. The minimal two-branch hard carrier family

Fix a transport window contained in

$$W_{\mathrm{tr}} \subseteq \{q : q \geq q_{\min} > 0\},$$

so that all branch derivatives are taken away from the cusp of the surrogate carrier dispersion. Then set

$$E_L(q) = \sqrt{m^2 + c_L^2 q^2}, \qquad E_T(q) = \delta + v_T|q|, \qquad v_T > c_L, \qquad \Delta(q) \equiv \Delta_0 > 0.$$

Then the explicit branch-transfer inverse kernel becomes

$$\mathcal{D}_\lambda(\omega, q) = \begin{pmatrix} \omega - E_L(q) & -\lambda\Delta_0 \\ -\lambda\Delta_0 & \omega - E_T(q) \end{pmatrix}, \qquad \lambda \in [0, 1].$$

This family does not pretend to be the final microphysics. It is the first explicit local model in which the transport algebra can be seen completely.

4.3.3 X.25.3. Exact branch spectrum

The eigenbranches are

$$E_\pm(q, \lambda) = \frac{E_L(q) + E_T(q)}{2} \pm \sqrt{\left(\frac{E_L(q) - E_T(q)}{2}\right)^2 + \lambda^2\Delta_0^2},$$

and the avoided-crossing gap is

$$\Delta_{\mathrm{gap}}(q, \lambda) = E_+(q, \lambda) - E_-(q, \lambda) = 2\sqrt{\left(\frac{E_L(q) - E_T(q)}{2}\right)^2 + \lambda^2\Delta_0^2}.$$

For $\lambda > 0$ the crossing is regularized and the transport path can be followed continuously.

4.3.4 X.25.4. The superluminal window

The carrier superluminal window is

$$U_T(\lambda) = (W_{\mathrm{tr}} \cap \{q : \ |\partial_q E_+(q, \lambda)| > c_L\}).$$

If $v_T > c_L$ and Δ_0 is sufficiently small relative to the branch separation outside the crossing zone, then $U_T(1) \neq \varnothing$. This is the first explicit spectral criterion for a hard carrier phase.

4.3.5 X.25.5. The crossing scale

The approximate crossing scale q_c is determined by

$$E_L(q_c) = E_T(q_c), \qquad \text{i.e.} \qquad \sqrt{m^2 + c_L^2 q_c^2} = \delta + v_T q_c.$$

The manuscript reads the hard transport picture as

$$\text{entry near the avoided crossing} \to \text{ride the carrier side} \to \text{return}.$$

The crossing scale is therefore both a kinematic and an operational quantity.

4.3.6 X.25.6. Adiabatic switching score

With $\lambda = \lambda(t)$, the natural adiabatic score is

$$\mathcal{A}(q,t) = \frac{|\dot{\lambda}(t)|\Delta_0}{\Delta_{\mathrm{gap}}(q,\lambda)^2}, \qquad \mathcal{A}_{\max} = \sup_{q,t} \mathcal{A}(q,t).$$

If $\mathcal{A}_{\max} \ll 1$, the selected transport branch can be followed adiabatically through the switching region. All derivative-based switching estimates are read on $U_T(\lambda) \cap W_{\mathrm{tr}}$.

4.3.7 X.25.7. First one-pole proxy

The first one-pole proxy records the explicit observability data that later feed the general projected-physical observability theorem. The one-pole proxy survives along the transfer if the selected branch remains simple, the branch gap never collapses,

$$\Delta_{\mathrm{gap}}(q,\lambda) > 0,$$

and the effective residue factor stays bounded away from zero and infinity,

$$0 < Z_{\min} \le Z_*(q,\lambda) \le Z_{\max} < \infty.$$

4.3.8 X.25.8. Identity-loss proxy

The first cycle fidelity proxy is

$$\mathcal{F}_{\mathrm{cyc}} := |\langle \psi_{\mathrm{in}}, \psi_{\mathrm{out}} \rangle|^2, \qquad \epsilon_{\mathrm{cyc}} := 1 - \mathcal{F}_{\mathrm{cyc}}.$$

This is weaker than the later structural hard identity metric, but it is enough for the first explicit one-pole model.

4.3.9 X.25.9. The first explicit one-pole transport model

The cycle is now explicit.

Input: a stable one-pole state near the luminal branch.

Stage 1: raise $\lambda(t)$ slowly from 0 to 1 while remaining near the selected upper branch.

Stage 2: transport the packet inside the nonempty superluminal window where

$$|\partial_q E_+(q,1)| > c_L.$$

Stage 3: lower $\lambda(t)$ back to 0 and reattach to the luminal branch.

Success requires

$$U_T(1) \neq \varnothing, \qquad \mathcal{A}_{\max} \ll 1, \qquad \epsilon_{\mathrm{cyc}} \ll 1, \qquad Z_* \text{ bounded}.$$

4.3.10 X.25.10. First explicit hard theorem

For the explicit two-branch family above, if the superluminal window is nonempty, the adiabatic score is small, the one-pole proxy survives, and the cycle identity-loss proxy is small, then the family realizes an explicit hard transport cycle with a superluminal carrier phase. This is the first closed explicit hard theorem of the branch-transfer family.

4.3.11 X.25.11. A concrete numerical instance

The manuscript fixes one explicit finite input:

$$m = 1, \qquad c_L = 1, \qquad \delta = 0.2, \qquad v_T = 1.8, \qquad \Delta_0 = 0.05.$$

This yields a genuinely superluminal carrier slope, a finite avoided-crossing gap for $\lambda > 0$, and a small-gap regime in which slow switching can produce $\mathcal{A}_{\max} \ll 1$ on a compact window. The point is not realism but calculability.

4.4 X.26. Projected Resolvent, Pole Survival, and the First Physical Hard Observability Criterion

4.4.1 X.26.1. Why projected resolvents are necessary

After X.25, the manuscript sharpens the hard question. A fast branch and an explicit switching family are not enough. The theory already distinguishes microscopic, sector-resolved, and physical observer resolvents. Physical observability must be read from the projected physical resolvent rather than from mere internal branch existence.

4.4.2 X.26.2. Data of the projected-resolvent language

Let $\mathcal{R}_{\mathrm{micro}}$ be the microscopic resolvent, $\mathcal{R}_\sigma$ the sector-resolved resolvent, and

$$\mathcal{R}_{\mathrm{phys}} = P_{\mathrm{phys}}\, \mathcal{R}_{\mathrm{micro}}\, P_{\mathrm{phys}}$$

the physical observer resolvent. For the transported object-sector $\sigma_{\mathfrak{p}}$, the relevant physical block is

$$\mathcal{R}^{(\mathfrak{p})}_{\mathrm{phys}} := P_{\sigma_{\mathfrak{p}}} \mathcal{R}_{\mathrm{phys}} P_{\sigma_{\mathfrak{p}}}.$$

The candidate carrier branch is physically realized only if near on-shell energy one has

$$\mathcal{R}^{(\mathfrak{p})}_{\mathrm{phys}}(\eta, q; \lambda) \sim \frac{Z_*(q,\lambda)}{\eta - E_*(q,\lambda)} + \mathcal{R}_{\mathrm{reg}}(\eta, q; \lambda),$$

with $Z_*(q,\lambda) \neq 0$.

4.4.3 X.26.3. The first physical hard observability criterion

The criterion is now explicit. The carrier phase at $\lambda = 1$ is physically observable only if:

1. the carrier window is nonempty,
2. the projected physical resolvent has an isolated simple pole at the selected branch,
3. the residue is nonzero on the relevant transport window,
4. the branch remains adiabatically accessible,
5. the cycle return remains close enough to the initial object-sector.

This is the first genuine bridge from a fast internal branch to a physically admissible hard carrier phase.

4.4.4 X.26.4. Necessary versus sufficient reading

If no isolated simple pole exists in the projected physical resolvent, the branch cannot count as a physical hard carrier. If the projected pole, nonzero residue, superluminal window, adiabatic accessibility, and small cycle loss all hold, then one has a candidate physical hard carrier phase. The manuscript insists on this distinction because admissibility at the internal or sector-resolved level is not yet enough.

4.4.5 X.26.5. Hard observability theorem [P]

If the branch-transfer family admits a selected branch with nonempty superluminal window, small adiabatic score, small cycle loss, and a projected physical resolvent block of the form

$$P_{\sigma_{\mathfrak{p}}} \mathcal{R}_{\mathrm{phys}} P_{\sigma_{\mathfrak{p}}} \sim \frac{Z_*(q,\lambda)}{\eta - E_*(q,\lambda)}$$

with $Z_* > 0$, then the family realizes a physically admissible one-pole hard carrier. This is the observability theorem for the explicit hard family.

4.4.6 X.26.6. First practical proxy for the projected pole

Before invoking the general projected-physical observability theorem of X.32.4, it is useful to introduce a practical proxy mask for observability:

$$\Pi_{\mathrm{hard}}(q,\lambda) := \mathbf{1}_{U_T}(q)\, \mathbf{1}_{\mathcal{A}_{\max} \ll 1}\, \mathbf{1}_{Z_*(q,\lambda) \in [Z_{\min}, Z_{\max}]}\, \mathbf{1}_{\epsilon_{\mathrm{cyc}} \ll 1}.$$

If $\Pi_{\mathrm{hard}}(q,\lambda) = 1$, the point is treated as a proxy-physical hard point for the explicit family. This proxy does not replace the general theorem. It is introduced only as an operational diagnostic at the explicit-model stage.

4.5 X.27. Pole Survival for the Explicit Hard Carrier Family

4.5.1 X.27.1. Why pole survival is the next bottleneck

The explicit family from X.25 and the projected criterion from X.26 reduce the next hard question to pole survival. The issue is no longer whether a fast branch exists, but whether the transported sector survives as a simple projected pole with nonzero residue along the path $L \to T \to L$.

4.5.2 X.27.2. The explicit family recalled

The family is

$$\mathcal{D}_\lambda(\omega, q) = \begin{pmatrix} \omega - E_L(q) & -\lambda\Delta_0 \\ -\lambda\Delta_0 & \omega - E_T(q) \end{pmatrix},$$

with

$$E_L(q) = \sqrt{m^2 + c_L^2 q^2}, \qquad E_T(q) = \delta + v_T|q|, \qquad q \in W_{\mathrm{tr}} \subseteq \{q : q \geq q_{\min} > 0\}.$$

Its eigenbranches are $E_\pm(q,\lambda)$, and for $\lambda > 0$ the gap $\Delta_{\mathrm{gap}}(q,\lambda)$ is strictly positive.

4.5.3 X.27.3. Sector-resolved and projected resolvents in the model

The full model resolvent is

$$\mathcal{R}_\lambda(\omega,q)=\mathcal{D}_\lambda(\omega,q)^{-1}=\frac{1}{\det\mathcal{D}_\lambda(\omega,q)}\begin{pmatrix}\omega-E_T(q) & \lambda\Delta_0\\ \lambda\Delta_0 & \omega-E_L(q)\end{pmatrix},$$

with

$$\det\mathcal{D}_\lambda(\omega,q)=(\omega-E_+(q,\lambda))(\omega-E_-(q,\lambda)).$$

This already shows that both eigenbranches appear as poles of the resolvent.

4.5.4 X.27.4. The luminal-sector projected resolvent

Projecting to the original luminal component gives

$$P_L=\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix},\qquad R_L:=P_L\mathcal{R}_\lambda P_L,$$

and hence

$$R_L(\omega,q;\lambda)=\frac{\omega-E_T(q)}{(\omega-E_+(q,\lambda))(\omega-E_-(q,\lambda))}=\left(\omega-E_L(q)-\frac{\lambda^2\Delta_0^2}{\omega-E_T(q)}\right)^{-1}.$$

The first form makes the poles explicit; the second makes the carrier-induced renormalization explicit.

4.5.5 X.27.5. Simplicity of the poles

For every $\lambda>0$ and every q, the poles of $R_L(\omega,q;\lambda)$ at $\omega=E_\pm(q,\lambda)$ are simple because the determinant factorizes into two distinct roots and the gap stays positive. This is the first exact pole-survival fact in the explicit hard family.

4.5.6 X.27.6. Residue factors

For the upper pole,

$$Z_+^{(L)}(q,\lambda)=\operatorname*{Res}_{\omega=E_+}R_L(\omega,q;\lambda)=\frac{E_+(q,\lambda)-E_T(q)}{E_+(q,\lambda)-E_-(q,\lambda)}.$$

Similarly,

$$Z_-^{(L)}(q,\lambda)=\frac{E_T(q)-E_-(q,\lambda)}{E_+(q,\lambda)-E_-(q,\lambda)}.$$

These formulas show that the residues are finite and nonzero as long as the denominator stays nonzero and the numerator does not collapse.

4.5.7 X.27.7. First nonvanishing-residue criterion

If the selected branch is E_+ and one has

$$E_+(q,\lambda)\neq E_T(q)\qquad\text{and}\qquad\Delta_{\mathrm{gap}}(q,\lambda)>0$$

on the transport window, then

$$Z_+^{(L)}(q,\lambda)\neq 0.$$

This is a purely model-level residue criterion, but it is exactly the kind of condition needed to approach the general projected-pole theorem.

4.5.8 X.27.8. The selected transported pole

The selected transported pole is not just any pole of the family. It is the continuous choice

$$(\omega, q, \lambda) \mapsto E_*(q, \lambda)$$

obtained by adiabatic continuation from the entry condition at $\lambda = 0$. Hard pole survival therefore means that the physically followed branch remains a simple projected pole with nonzero residue throughout the full path.

4.5.9 X.27.9. Pole survival theorem for the explicit family

For the explicit two-branch hard carrier family, every selected transported branch continuously followed through the avoided-crossing region remains a simple pole of the sector-resolved projected resolvent as long as:

1. $\lambda > 0$,
2. $\Delta_{\mathrm{gap}}(q, \lambda) > 0$,
3. the corresponding residue numerator does not vanish on the transport window.

This is already an exact pole-survival statement for the sector-resolved model. The next closure step is the lift from this sector-resolved pole to the full physical projected pole of the observer channel.

4.5.10 X.27.10. From sector-resolved pole to physical pole

The lift from a sector-resolved pole to a physical pole requires more than the abstract phrase "physical admissibility." The manuscript fixes three explicit hypotheses on the working window: the physical projector does not annihilate the selected transported sector,

$$P_{\mathrm{phys}}\Pi_* \neq 0,$$

the projected block preserves pole isolation and nonzero residue under the chosen observer reduction, and admissible perturbations do not change the projector class. Under these three conditions, a simple nonzero-residue pole of the sector-resolved block survives as a pole of the selected object-sector block of the physical projected resolvent. This is the physical lifting step of the hard theorem chain.

4.5.11 X.27.11. First hard physical pole corollary

If the explicit family satisfies sector-resolved pole survival, non-annihilation by P_{phys}, projection compatibility on the working observer window, and a nonempty superluminal carrier window, then the family admits a physically observable superluminal one-pole carrier phase in the selected object-sector block on that window. This is one of the strongest exact results of the current hard chain.

4.6 X.28. Identity Preservation for the Explicit Hard Carrier Cycle

4.6.1 X.28.1. Why identity is the next barrier

After X.27, the explicit family can preserve a projected pole. But hard FTL requires more than the survival of some pole. It requires the return of the same object-sector after the cycle. The

manuscript therefore asks not merely whether a pole survives, but whether the transported sector remains the same object-sector after

$$L \to T \to L.$$

4.6.2 X.28.2. What "the same object" means here

The manuscript distinguishes three levels of identity:

pole identity, meaning that the returned sector still has the same asymptotic one-pole type;

residue identity, meaning that residue factors and attached observables drift only slightly;

transport identity, meaning that the returned object lies in the same observer-level transport class rather than being destroyed and reassembled into a different one.

4.6.3 X.28.3. The cycle for the explicit family

For the explicit family one has the exact loop

$$\lambda : 0 \to 1 \to 0.$$

At the input the object sits on the luminal branch. In the middle it travels along the selected branch. At the output the family returns to the original inverse kernel, so a before-versus-after comparison is meaningful.

4.6.4 X.28.4. Minimal identity metrics

The manuscript introduces the first explicit hard identity metrics:

$$\mathcal{F}_{\mathrm{cyc}} := |\langle \psi_{\mathrm{in}}, \psi_{\mathrm{out}} \rangle|^2, \qquad \epsilon_{\mathrm{st}} := 1 - \mathcal{F}_{\mathrm{cyc}},$$

$$\epsilon_Z := \sup_{q \in U_{\mathrm{ret}}} \frac{|Z_{\mathrm{out}}(q) - Z_{\mathrm{in}}(q)|}{|Z_{\mathrm{in}}(q)|}, \qquad \epsilon_Q := \sum_a |Q_a^{\mathrm{out}} - Q_a^{\mathrm{in}}|,$$

and then combines them into

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} = w_{\mathrm{st}} \epsilon_{\mathrm{st}} + w_Z \epsilon_Z + w_Q \epsilon_Q.$$

This is already stronger than a bare overlap proxy.

4.6.5 X.28.5. Ideal adiabatic return

In the ideal adiabatic and nondissipative limit, if the selected branch remains simple and isolated and the family returns exactly to its initial form at $\lambda = 0$, then one expects

$$\psi_{\mathrm{out}} = e^{i\varphi} \psi_{\mathrm{in}}, \qquad \mathcal{F}_{\mathrm{cyc}} = 1, \qquad \epsilon_{\mathrm{st}} = 0.$$

The manuscript emphasizes that cycle identity loss is therefore not forced by the mere structure of the hard loop. The barriers are leakage, residue drift, and loss of observability.

4.6.6 X.28.6. First estimate of identity loss

If the nonadiabatic leakage is bounded by L_{nad}, the residue drift by ϵ_Z, and the observable drift by ϵ_Q, then one has the first control estimate

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} \lesssim w_{\mathrm{st}} L_{\mathrm{nad}} + w_Z \epsilon_Z + w_Q \epsilon_Q.$$

So if

$$\mathcal{A}_{\max} \ll 1, \qquad \epsilon_Z \ll 1, \qquad \epsilon_Q \ll 1,$$

then

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} \ll 1.$$

This is the first local hard identity-control statement.

4.6.7 X.28.7. Relation to pole survival

The manuscript insists that X.27 and X.28 close different barriers. One may have a surviving projected pole but large cycle identity loss, or a good return overlap but no physically admissible pole in the carrier phase. Hard FTL requires both:

$$\boxed{\text{pole survival} + \text{identity preservation.}}$$

4.6.8 X.28.8. First theorem for an identity-preserving hard cycle [P]

If the explicit branch-transfer family satisfies sector-resolved pole survival on the selected branch, physical admissibility of the transported one-pole sector, adiabatic accessibility, small residue drift, and small conserved-observable drift, then the full cycle

$$\mathfrak{p}_L \to \mathfrak{p}_T \to \mathfrak{p}_L$$

is identity-preserving at the first hard observer-relevant level:

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} \ll 1.$$

This is the first closed identity theorem for the explicit hard cycle.

4.7 X.29. Binding Stability in the Carrier Phase and the First Composite Hard Criterion

4.7.1 X.29.1. Why composite hard transport is a separate barrier

After X.28, the hard route has a one-pole carrier cycle with pole survival and an identity theorem. But that still does not amount to transport of a bound physical object. Composite sectors must be treated separately. The manuscript therefore asks whether a bound object can survive inside the carrier phase.

4.7.2 X.29.2. What the theory already gives for bound states

The general framework already supplies the necessary language: bound states are energetically stable when their joint rest energy lies below the sum of the parts, and chemistry-level composite sectors are physically admissible only if their reduced composite resolvent has a stable isolated pole or at least a narrow resonance branch. That is the exact pattern that must now be transported into the hard carrier setting.

4.7.3 X.29.3. Carrier-phase composite sector

Let $\mathfrak{p}_1, \dots, \mathfrak{p}_n$ be constituent one-pole sectors that each admit a hard carrier cycle. Let

$$\mathfrak{P}$$

be a composite sector assembled from them by an admissible binding channel. The manuscript says that $\mathfrak{P}$ admits a carrier-phase composite image if there exists a family of composite inverse kernels

$$D_{\mathfrak{P},\lambda}(E, q)$$

whose reduced resolvent

$$G_{\mathfrak{P},\lambda}(E, q) := D_{\mathfrak{P},\lambda}(E, q)^{-1}$$

contains a selected composite branch

$$E_{\mathfrak{P},*}(q, \lambda)$$

that continuously deforms the ordinary bound branch into a carrier-phase bound branch.

4.7.4 X.29.4. Binding margin in the carrier phase

Let $E_{\mathrm{sep}}^{(T)}(q, \lambda)$ be the breakup threshold in the carrier phase. For two constituents one writes

$$E_{\mathrm{sep}}^{(T)}(q, \lambda) = \inf_{q_1+q_2=q} \big(E_{1,*}(q_1, \lambda) + E_{2,*}(q_2, \lambda)\big).$$

Then define the carrier-phase binding margin

$$B_T(q, \lambda) := E_{\mathrm{sep}}^{(T)}(q, \lambda) - E_{\mathfrak{P},*}(q, \lambda).$$

If $B_T(q, \lambda) > 0$, the composite branch lies below breakup threshold and is bound inside the carrier phase.

4.7.5 X.29.5. Composite pole criterion in the carrier phase

The manuscript then imposes the composite pole condition. The composite carrier branch survives physically if the reduced composite resolvent has an isolated pole

$$G_{\mathfrak{P},\lambda}(E, q) \sim \frac{Z_{\mathfrak{P}}(q, \lambda)}{E - E_{\mathfrak{P},*}(q, \lambda)}, \qquad Z_{\mathfrak{P}}(q, \lambda) \neq 0,$$

or at least a sufficiently narrow resonance branch.

4.7.6 X.29.6. The first composite hard criterion

Composite hard transport is candidate-admissible only if:

1. each constituent sector admits the one-pole carrier cycle;
2. the carrier-phase binding margin is positive;
3. the composite reduced resolvent has an isolated simple pole or narrow resonance;
4. the composite cycle identity loss is small;
5. the composite branch itself is superluminal on a nonempty window.

Only the conjunction of these five items yields a plausible composite hard transport regime.

4.7.7 X.29.7. The first composite hard theorem [P]

If each constituent one-pole sector satisfies the one-pole hard theorem, the composite carrier branch exists, the binding margin is positive on the carrier window, the composite reduced resolvent has an isolated simple pole or narrow resonance, and the composite cycle identity loss is small, then the composite object realizes a composite hard transport phase. This is the first closed composite theorem of the explicit hard package.

4.7.8 X.29.8. The explicit two-constituent prototype

To make the criterion concrete, the manuscript introduces a two-constituent prototype with branches

$$E_{1,*}(q_1, \lambda), \qquad E_{2,*}(q_2, \lambda),$$

and carrier-phase composite branch

$$E_{\mathfrak{P},*}(q, \lambda) = \inf_{q_1+q_2=q} \big(E_{1,*}(q_1, \lambda) + E_{2,*}(q_2, \lambda)\big) - \mathcal{B}(q, \lambda), \qquad \mathcal{B}(q, \lambda) \geq 0.$$

Then

$$B_T(q, \lambda) = \mathcal{B}(q, \lambda),$$

so carrier-phase binding reduces to the positivity of the effective binding gain.

4.8 X.30. Explicit Composite Hard Carrier Model and the First Bound-Transport Example

4.8.1 X.30.1. Why the first composite model must be minimal

The manuscript now does for composite transport what X.25 did for one-pole transport. It takes the smallest explicit model in which one can calculate a binding margin, a composite branch, a composite pole, and a composite identity-loss proxy.

4.8.2 X.30.2. The two-constituent carrier prototype

Take two constituent one-pole sectors that each admit the explicit hard cycle from X.25–X.28. In the symmetric case,

$$E_{1,*}(q, \lambda) = E_{2,*}(q, \lambda) = E_*(q, \lambda).$$

Define the composite branch by

$$E_{\mathfrak{P},*}(Q, \lambda) = \inf_{q_1+q_2=Q} \big(E_*(q_1, \lambda) + E_*(q_2, \lambda)\big) - \mathcal{B}(Q, \lambda), \qquad \mathcal{B}(Q, \lambda) \geq 0.$$

4.8.3 X.30.3. Separation threshold and binding margin

The separation threshold is

$$E^{(T)}_{\mathrm{sep}}(Q, \lambda) = \inf_{q_1+q_2=Q} \big(E_*(q_1, \lambda) + E_*(q_2, \lambda)\big),$$

so the binding margin becomes

$$B_T(Q, \lambda) = E^{(T)}_{\mathrm{sep}}(Q, \lambda) - E_{\mathfrak{P},*}(Q, \lambda) = \mathcal{B}(Q, \lambda).$$

Thus the simplest explicit composite hard criterion is just

$$\boxed{B_T(Q, \lambda) > 0 \iff \mathcal{B}(Q, \lambda) > 0.}$$

4.8.4 X.30.4. Composite inverse kernel and composite resolvent

The minimal composite inverse kernel is

$$D_{\mathfrak{P},\lambda}(E,Q) = E - E_{\mathfrak{P},*}(Q,\lambda),$$

with reduced composite resolvent

$$G_{\mathfrak{P},\lambda}(E,Q) = \frac{1}{E - E_{\mathfrak{P},*}(Q,\lambda)}.$$

If one wishes to include decay, one may instead write

$$D_{\mathfrak{P},\lambda}(E,Q) = E - E_{\mathfrak{P},*}(Q,\lambda) + i\Gamma_{\mathfrak{P}}(Q,\lambda),$$

so that $\Gamma_{\mathfrak{P}} = 0$ gives an isolated pole and small positive $\Gamma_{\mathfrak{P}}$ gives a narrow resonance branch.

4.8.5 X.30.5. First explicit composite admissibility criterion

The explicit two-constituent carrier prototype is composite-hard-admissible on a window $W_{\mathfrak{P}}$ if

$$B_T(Q,\lambda) = \mathcal{B}(Q,\lambda) > 0,$$

the composite width is zero or sufficiently small,

$$\Gamma_{\mathfrak{P}}(Q,\lambda) = 0 \quad \text{or} \quad \Gamma_{\mathfrak{P}}(Q,\lambda) \ll 1,$$

the composite branch outruns the luminal reference on a nonempty subwindow, and the constituent one-pole hard conditions remain valid.

4.8.6 X.30.6. The first bound-transport example

Take the carrier-phase constituent branch in the working window to be approximately

$$E_*(q,1) \approx \delta + v_T|q|, \qquad v_T > c_L.$$

For total momentum $Q \geq 0$ one then gets

$$E^{(T)}_{\mathrm{sep}}(Q,1) = 2\delta + v_T Q.$$

With constant binding gain

$$\mathcal{B}(Q,1) \equiv B_0 > 0,$$

the composite branch is

$$E_{\mathfrak{P},*}(Q,1) = 2\delta + v_T Q - B_0,$$

and hence

$$B_T(Q,1) = B_0 > 0, \qquad |\partial_Q E_{\mathfrak{P},*}(Q,1)| = v_T > c_L.$$

The minimal composite resolvent is then

$$G_{\mathfrak{P},1}(E,Q) = \frac{1}{E - (2\delta + v_T Q - B_0)},$$

which has an isolated simple pole with residue 1. All composite slope statements are read on the working window away from the surrogate cusp inherited from the constituent carrier branch.

4.8.7 X.30.7. First explicit composite theorem

For the explicit two-constituent carrier prototype with

$$E_*(q,1) = \delta + v_T|q|, \qquad v_T > c_L, \qquad \mathcal{B}(Q,1) = B_0 > 0,$$

the composite branch satisfies positive binding margin, isolated simple composite pole, and composite superluminality. Hence the model realizes the first explicit bound-transport example inside the closed hard theorem chain.

4.8.8 X.30.8. What this example really means

The example does not prove that any realistic macroscopic body can be transported hard-FTL. It proves the weaker but nontrivial claim that the theory admits an explicit composite carrier-phase model in which binding, pole survival, and superluminal transport coexist.

4.8.9 X.30.9. Composite identity-loss proxy

For the full cycle

$$\mathfrak{P}_L \to \mathfrak{P}_T \to \mathfrak{P}_L,$$

the manuscript adds a binding-drift term to the composite identity functional:

$$\epsilon_B^{\mathrm{comp}} := \sup_Q \frac{|B_{\mathrm{out}}(Q) - B_{\mathrm{in}}(Q)|}{|B_{\mathrm{in}}(Q)|},$$

and then defines

$$\epsilon_{\mathrm{id}}^{\mathrm{comp}} = w_{\mathrm{st}}\epsilon_{\mathrm{st}}^{\mathrm{comp}} + w_Z\epsilon_Z^{\mathrm{comp}} + w_Q\epsilon_Q^{\mathrm{comp}} + w_B\epsilon_B^{\mathrm{comp}}.$$

For a composite object, one must return not only the same constituents but the same bound structure.

4.9 X.31. Robustness of One-Pole and Composite Hard Transport and the Hard-Closure Problem

4.9.1 X.31.1. Why hard transport now needs a robustness audit

After X.30, the hard route already has an explicit one-pole carrier family, projected pole survival for the explicit family, a one-pole identity theorem, and an explicit composite bound-transport prototype. The next step is therefore a robustness audit. The branch should survive admissible perturbations as a package rather than only in one fine-tuned family.

4.9.2 X.31.2. The right notion of robustness in this framework

Hard robustness means more than small formula drift. It means that the superluminal window remains nonempty, the projected pole remains simple and nonzero-residue, the selected branch remains adiabatically accessible, the cycle identity loss remains small, and for composites the binding margin stays positive while the composite pole survives.

4.9.3 X.31.3. Admissible perturbation class for hard transport

An admissible hard perturbation consists of small deformations

$$\delta\mathcal{D}_\lambda, \qquad \delta P_{\mathrm{phys}}, \qquad \delta\lambda(t), \qquad \delta D_{\mathfrak{P},\lambda},$$

such that the observer regime stays fixed, the reference luminal sector remains fixed, the selected transported branch remains continuously identifiable, and the physical projector class does not jump into a different superselection type.

Fix compact working windows

$$W_{\mathrm{tr}} \subset (q_{\min}, \infty) \times [0,1], \qquad W_P$$

for the one-pole and composite sectors, respectively. The manuscript works inside the following admissible hard perturbation class.

Definition 4.5 Admissible hard perturbation class. An admissible hard perturbation is a quadruple

$$(\delta D_\lambda, \delta P_{\mathrm{phys}}, \delta\lambda, \delta D_{P,\lambda})$$

such that

1. $\delta D_\lambda \in C^1(W_{\mathrm{tr}}; M_2(\mathbb{C}))$,
2. $\delta P_{\mathrm{phys}} \in C^0(W_{\mathrm{tr}}; \mathcal{B}(\mathcal{H}_{\mathrm{phys}}))$,
3. $\delta\lambda \in C^1([0,T])$,
4. $\delta D_{P,\lambda} \in C^1(W_P; \mathbb{C})$,

and the selected one-pole branches $E_*(q,\lambda)$, residues $Z_*(q,\lambda)$, composite branches $E_{P,*}(Q,\lambda)$, carrier-phase binding margins $B_T(Q,\lambda)$, and projector-overlap map

$$(q,\lambda) \mapsto P_{\mathrm{phys}}\Pi_*(q,\lambda)$$

all depend continuously on the perturbation in the norm fixed below.

For the general hard closure the manuscript fixes the perturbation norm

$$\|\delta\|_{\mathrm{hard}} := \sup_{W_{\mathrm{tr}}} \|\delta D_\lambda\|_{C^1} + \sup_{W_{\mathrm{tr}}} \|\delta P_{\mathrm{phys}}\|_{\mathrm{op}} + \|\delta\lambda\|_{C^1([0,T])} + \sup_{W_P} \|\delta D_{P,\lambda}\|_{C^1}.$$

All general theorems H-C1 through H-C5 are read relative to this fixed norm and a fixed admissible class in which the observer regime, reference luminal sector, and projector class remain unchanged. All continuity and openness statements in Lemmas 4.6 through 4.10 and Theorems H-C1 through H-C5 are understood relative to this perturbation class.

4.9.4 X.31.4. One-pole robustness margins

The manuscript defines four one-pole margins:

$$M_{\mathrm{vel}} := \inf_{q\in U_T} (|\partial_q E_*(q,1)| - c_L),$$

$$M_{\mathrm{gap}} := \inf_{(q,\lambda)\in W_{\mathrm{tr}}} \Delta_{\mathrm{gap}}(q,\lambda),$$

$$M_Z := \inf_{(q,\lambda)\in W_{\mathrm{tr}}} |Z_*(q,\lambda)|,$$

$$M_{\mathrm{id}} := 1 - \epsilon_{\mathrm{id}}^{\mathrm{cyc}}.$$

These margins protect speed, branch isolation, observable pole weight, and object identity.

4.9.5 X.31.5. One-pole robustness load

The perturbation load is summarized by

$$R_{1\mathrm{p}} := \zeta_{\mathrm{br}} + \zeta_Z + L_{\mathrm{nad}} + \zeta_{\mathrm{id}},$$

where the terms bound branch-speed and gap drift, residue drift, nonadiabatic leakage, and induced identity degradation. If

$$R_{1\mathrm{p}} < \min\{M_{\mathrm{vel}}, M_{\mathrm{gap}}, M_Z, M_{\mathrm{id}}\},$$

then one-pole hard transport survives admissible perturbation as a physically observable candidate cycle.

4.9.6 X.31.6. Composite robustness margins

For composite transport one adds

$$\begin{aligned} M_{\mathrm{bind}} &:= \inf_{(Q,\lambda)\in W_{\mathfrak{P}}} B_T(Q,\lambda), \\ M_{\mathrm{comp}Z} &:= \inf_{(Q,\lambda)\in W_{\mathfrak{P}}} |Z_{\mathfrak{P}}(Q,\lambda)|, \\ M_\Gamma &:= \frac{1}{10} - \sup_{(Q,\lambda)\in W_{\mathfrak{P}}} \Gamma_{\mathfrak{P}}(Q,\lambda), \\ M_{\mathrm{compid}} &:= 1 - \epsilon_{\mathrm{id}}^{\mathrm{comp}}. \end{aligned}$$

These margins protect boundness, composite observability, resonance narrowness, and composite sameness.

4.9.7 X.31.7. Composite robustness load

The composite perturbation load is

$$R_{\mathrm{comp}} := \zeta_{\mathrm{bind}} + \zeta_{\mathrm{comp}Z} + L_{\mathrm{nad}}^{\mathrm{comp}} + \zeta_{\mathrm{compid}}.$$

If

$$R_{\mathrm{comp}} < \min\{M_{\mathrm{bind}}, M_{\mathrm{comp}Z}, M_\Gamma, M_{\mathrm{compid}}\},$$

then the explicit composite hard carrier model remains a candidate bound-transport phase under admissible perturbations.

4.9.8 X.31.8. The hard robustness theorem [P]

The manuscript then combines the one-pole and composite parts. If the explicit hard package has a one-pole branch-transfer carrier family, projected simple-pole survival, small one-pole cycle identity loss, and an explicit composite carrier model with positive binding margin and composite pole, and if both robustness loads stay below the corresponding positive margins, then hard transport survives as a robust theorem package at both the one-pole and composite levels.

4.9.9 X.31.9. Perturbative lemmas for general hard closure

The general hard theorems require five quantitative perturbative lemmas. Each one is stated relative to the fixed admissible class and the norm $\|\delta\|_{\mathrm{hard}}$.

Lemma 4.6 [P]. Openness of the nonempty carrier window. *Fix an explicit branch-transfer family with positive velocity margin $M_{\mathrm{vel}} > 0$ on a transport window U_T, within the admissible hard perturbation class of Definition 4.5. Then there exists $\varepsilon_{\mathrm{vel}} > 0$ such that every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{vel}}$ preserves a nonempty subwindow on which*

$$|\partial_q E_T^{\delta}(q)| > c_L.$$

In particular, the nonempty superluminal carrier window is an open condition.

Lemma 4.7 [P]. Openness of simple-pole and nonzero-residue data. *Fix a selected transported branch with positive pole-isolation and residue margins $M_{\mathrm{gap}} > 0$ and $M_Z > 0$ on W_{tr}, within the admissible hard perturbation class of Definition 4.5. Then there exists $\varepsilon_{\mathrm{pole}} > 0$ such that every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{pole}}$ preserves both simplicity of the selected projected pole and nonvanishing of its residue on the working window.*

Lemma 4.8 [P]. Continuity of the one-pole identity functional. *Fix a one-pole cycle with $\epsilon_{\mathrm{id}}^{\mathrm{cyc}} < 1$ and positive identity margin $M_{\mathrm{id}} > 0$, within the admissible hard perturbation class of Definition 4.5. Then there exists a constant $C_{\mathrm{id}} > 0$ and a radius $\varepsilon_{\mathrm{id}} > 0$ such that for every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{id}}$ one has*

$$|\epsilon_{\mathrm{id}}^{\mathrm{cyc},\delta} - \epsilon_{\mathrm{id}}^{\mathrm{cyc}}| \le C_{\mathrm{id}} \|\delta\|_{\mathrm{hard}},$$

and therefore $\epsilon_{\mathrm{id}}^{\mathrm{cyc},\delta} < 1$ whenever $C_{\mathrm{id}}\|\delta\|_{\mathrm{hard}} < M_{\mathrm{id}}$.

Lemma 4.9 [P]. Openness of positive binding margin, narrow resonance, and composite identity compatibility. *Fix a composite branch with positive binding, composite-residue, narrow-resonance, and composite-identity margins*

$$M_{\mathrm{bind}} > 0, \qquad M_{\mathrm{comp}Z} > 0, \qquad M_{\Gamma} > 0, \qquad M_{\mathrm{compid}} > 0$$

on $W_{\mathfrak{P}}$, within the admissible hard perturbation class of Definition 4.5.

Then there exist constants $C_{\mathrm{comp}} > 0$, $C_{\mathrm{compid}} > 0$, and a radius $\varepsilon_{\mathrm{comp}} > 0$ such that every admissible perturbation with

$$\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{comp}}$$

satisfies

$$|B_T^{\delta} - B_T| + |Z_{\mathfrak{P}}^{\delta} - Z_{\mathfrak{P}}| + |\Gamma_{\mathfrak{P}}^{\delta} - \Gamma_{\mathfrak{P}}| \le C_{\mathrm{comp}} \|\delta\|_{\mathrm{hard}},$$

and

$$|\epsilon_{\mathrm{id}}^{\mathrm{comp},\delta} - \epsilon_{\mathrm{id}}^{\mathrm{comp}}| \le C_{\mathrm{compid}} \|\delta\|_{\mathrm{hard}}.$$

Hence, if $\varepsilon_{\mathrm{comp}}$ is chosen so that

$$C_{\mathrm{comp}}\|\delta\|_{\mathrm{hard}} < \min\{M_{\mathrm{bind}}, M_{\mathrm{comp}Z}, M_{\Gamma}\}, \qquad C_{\mathrm{compid}}\|\delta\|_{\mathrm{hard}} < M_{\mathrm{compid}},$$

then positive carrier-phase binding, composite observability, and composite identity compatibility all survive.

Lemma 4.10 [P]. Physical projection non-annihilation and class stability. *Fix a selected transported sector with projector overlap margin*

$$\kappa_{\mathrm{phys}} := \inf_{W_{\mathrm{tr}}} \|P_{\mathrm{phys}}\Pi_*\|_{\mathrm{op}} > 0.$$

Assume also that the physical projected block preserves pole isolation on W_{tr} and that admissible perturbations keep the projector class fixed, within the admissible hard perturbation class of Definition 4.5. Then there exists $\varepsilon_{\mathrm{phys}} > 0$ such that every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{phys}}$ preserves non-annihilation by P_{phys} and keeps the selected transported sector inside the same physical projection class.

4.9.10 X.31.10. The hard-closure problem

Hard closure means the conjunction of six items:

1. carrier existence,
2. one-pole observability,
3. one-pole identity preservation,
4. composite binding,
5. composite observability and identity compatibility,
6. robustness of the whole one-pole-plus-composite package.

This is the hard analogue of the soft closure statement.

4.9.11 X.31.11. The audited status of hard closure

Hard closure is not experimental realization, but it is theorem closure. At this stage the hard branch is closed as an audited open-class theorem package. What already exists is an explicit one-pole carrier family, projected-pole survival for the explicit family, a one-pole identity theorem, an explicit composite carrier prototype, positive one-pole and composite robustness margins, and the perturbative lemmas that turn those explicit margins into the general hard-closure theorems.

4.9.12 X.31.12. What can already be honestly claimed

What can already be claimed is an audited hard transport program consisting of a well-posed one-pole carrier problem, an explicit projected-pole survival result for a two-branch family, an identity-preserving cycle theorem, an explicit composite bound-transport prototype, and open-class stability for both one-pole and composite hard transport.

Proposition 4.11 [P]. The hard ladder is constrained by identity and binding observability. *Every advancement in the hard prototype ladder must preserve the earlier observability package while strengthening either the admissible carrier class or the transport complexity. In particular, a later composite stage is acceptable only if it retains the one-pole identity discipline and adds controlled binding observability rather than replacing identity control with mere carrier speed.*

This proposition is the formal expression of the manuscript order: first a one-pole admissible carrier cycle, then a composite carrier cycle with explicit binding criteria, and then open-class lifting of the whole hard package.

4.10 X.32. Package-Level Hard Closure and the General Hard Theorems

4.10.1 X.32.1. Why hard closure is now a completed chain

One of the main achievements of the completed hard route is that the theorem chain now closes end-to-end. The explicit-family results of X.24 through X.31 are not isolated calculations; together with the perturbative lemmas of X.31.9 and the closure theorems stated below, they close the hard route as a finite theorem chain rather than as an open-ended program.

4.10.2 X.32.2. What the completed hard chain contains

The route now contains transport sectors, pole-residue language, an explicit two-branch carrier family, quantitative projection-compatibility hypotheses, explicit and general identity theorems, an explicit composite carrier prototype, robustness estimates, and the general hard-closure theorems that place the hard package inside an admissible open class. The hard branch is therefore closed at theorem level, even though later calibration and realization questions still remain.

4.10.3 X.32.3. Theorem H-C1 [P]: general carrier existence

Fix the admissible class determined in X.31.3, within the admissible hard perturbation class of Definition 4.5, and the perturbation norm $\|\delta\|_{\mathrm{hard}}$. If the explicit branch-transfer family has positive velocity margin $M_{\mathrm{vel}} > 0$, then by Lemma 4.6 there exists $\varepsilon_{\mathrm{vel}} > 0$ such that every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{vel}}$ preserves a nonempty transport subwindow on which

$$|\partial_q E_T^{\delta}(q)| > c_L.$$

Hence the explicit branch-transfer family extends to a nontrivial admissible class of transport sectors with a nonempty superluminal carrier window. This is the general carrier-existence theorem H-C1.

4.10.4 X.32.4. Theorem H-C2 [P]: general projected-physical observability

Fix the same admissible class, within the admissible hard perturbation class of Definition 4.5, and the perturbation norm. Assume the selected transported branch survives as a simple nonzero-residue sector-resolved pole with margins $M_{\mathrm{gap}} > 0$ and $M_Z > 0$, assume the physical overlap margin $\kappa_{\mathrm{phys}} > 0$, and assume the projector class is stable on the working window. Then Lemma 4.7 and Lemma 4.10 give a radius $\varepsilon_{\mathrm{phys}} > 0$ such that every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{phys}}$ preserves the physical asymptotic pole of the selected object-sector block

$$P_{\sigma_p} P_{\mathrm{phys}} \mathcal{R}_{\mathrm{micro}} P_{\mathrm{phys}} P_{\sigma_p} \sim \frac{Z_*^{\delta}(q)}{\eta - E_*^{\delta}(q)}.$$

This is the indispensable bridge from internal carrier structure to physical hard transport, and it is part of the completed hard closure package.

4.10.5 X.32.5. Theorem H-C3 [P]: general identity preservation

Fix the same admissible class, within the admissible hard perturbation class of Definition 4.5, and assume the explicit one-pole cycle satisfies $\epsilon_{\mathrm{id}}^{\mathrm{cyc}} < 1$ with positive identity margin $M_{\mathrm{id}} > 0$. Then Lemma 4.8 provides $C_{\mathrm{id}} > 0$ and $\varepsilon_{\mathrm{id}} > 0$ such that every admissible perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{id}}$ and $C_{\mathrm{id}}\|\delta\|_{\mathrm{hard}} < M_{\mathrm{id}}$ preserves a switching regime in which the full cycle

$$\mathfrak{p}_L \to \mathfrak{p}_T \to \mathfrak{p}_L$$

preserves the one-pole object-sector in the strong observer-relevant sense,

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc},\delta} < 1.$$

Pole survival alone does not imply this, which is why H-C3 is a separate step in the completed closure chain.

4.10.6 X.32.6. Theorem H-C4 [P]: general composite observability, binding, and identity compatibility

Fix the same admissible class, within the admissible hard perturbation class of Definition 4.5, and assume the explicit composite branch has positive binding and observability margins $M_{\mathrm{bind}} > 0$, $M_{\mathrm{comp}Z} > 0$, and $M_\Gamma > 0$, together with a positive composite identity margin

$$M_{\mathrm{compid}} := 1 - \epsilon_{\mathrm{id}}^{\mathrm{comp}} > 0.$$

Assume also that the constituent one-pole cycles remain inside the admissible one-pole class. Then Lemma 4.9 gives $\varepsilon_{\mathrm{comp}} > 0$ such that every admissible perturbation with

$$\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{comp}}$$

preserves positive carrier-phase binding,

$$B_T^\delta(Q,\lambda) > 0,$$

and the reduced composite resolvent carries either an isolated pole or a sufficiently narrow resonance branch,

$$G_{\mathfrak{P},\lambda}(E,Q) \sim \frac{Z_{\mathfrak{P}}^\delta(Q,\lambda)}{E - E_{\mathfrak{P},*}^\delta(Q,\lambda)},$$

and the composite cycle identity loss remains below the admissible threshold

$$\epsilon_{\mathrm{id}}^{\mathrm{comp},\delta} < 1,$$

while the composite branch itself outruns the luminal reference on a nonempty window. This is the general composite observability, binding, and identity-compatibility theorem H-C4.

4.10.7 X.32.7. Theorem H-C5 [P]: hard robustness lifting

The fifth theorem lifts the explicit prototype to an admissible structural class, within the admissible hard perturbation class of Definition 4.5. If the hard package has positive one-pole and composite margins and the perturbation loads satisfy

$$R_{1\mathrm{p}} < \min\{M_{\mathrm{vel}}, M_{\mathrm{gap}}, M_Z, M_{\mathrm{id}}\}, \qquad R_{\mathrm{comp}} < \min\{M_{\mathrm{bind}}, M_{\mathrm{comp}Z}, M_\Gamma, M_{\mathrm{compid}}\},$$

then there exists an admissible perturbative neighborhood in which the superluminal window survives, projected poles survive, identity losses remain small, composite binding remains positive, and the composite transported sector remains physically observable. This is the robustness lifting theorem H-C5.

4.10.8 X.32.8. Dependency order of the hard chain

The correct order is

$$H\text{-}C1 \to H\text{-}C2 \to H\text{-}C3 \to H\text{-}C4 \to H\text{-}C5.$$

General carrier existence comes first, then the projected-physical bridge, then one-pole identity preservation, then composite observability, binding, and identity compatibility, and finally robustness lifting.

4.10.9 X.32.9. The hard-closure statement

Hard closure is achieved within the fixed admissible perturbation class because the five statements H-C1 through H-C5 hold as one dependency-ordered conjunction: general carrier existence, general projected physical observability, general one-pole identity preservation, general composite binding, observability, and identity compatibility, and robustness lifting to an admissible neighborhood. Chapter 6 proves these steps in the same dependency order using the perturbation norm, positivity margins, openness lemmas, continuity estimates, and projection-compatibility hypotheses fixed above. This is theorem-level closure of the hard route under the explicit admissibility and perturbative hypotheses of the manuscript, not a claim of unrestricted empirical realization.

With the explicit hard theorem chain and the corresponding general closure theorems stated, the hard route has reached its correct audit boundary inside the present manuscript.

5 Integrated Soft-Hard Architecture

This chapter isolates the minimal integrated layer after separate soft and hard closure. It begins with the audit boundary between route closure and pre-engineering closure, states the minimal soft-hard composition law, and then records the first integrated target $I0$. The purpose is to keep the theory line focused on composition, not to absorb the later engineering notebooks into the theorem core.

The key ordering principle is sharp. Soft and hard branches must first be closed separately at the theorem-package level. Only after that may one speak about integration. Soft FTL is memory- and readout-dominated; hard FTL is carrier- and transport-dominated. If they are fused too early, the result becomes rhetorically ambitious but mathematically unclear. The integrated architecture must therefore begin with the smallest composition law that preserves both route identities.

5.1 X.45. Final Audit and the Need for a Separate Integration Layer

5.1.1 X.45.1. What is already theoretically closed

The manuscript first states what is already closed before integration is attempted.

Soft FTL is already closed as a theorem package consisting of a branch-locked soft sector, a binary witness, a finite scanner, a hierarchical remote-scene approximation, a detector-plus-memory operational protocol, and a robustness theorem with a soft-closure statement.

Hard FTL is already closed as a theorem package consisting of explicit one-pole carrier families, projected-pole survival for the explicit family, identity-preserving carrier cycles, explicit composite bound-transport prototypes, quantitative one-pole and composite robustness margins, and the general hard closure chain H-C1 through H-C5 proved from the perturbative lemmas of the hard route.

This matters for integration because the first integrated layer must consume already closed route packages rather than trying to stabilize them while composing them.

5.1.2 X.45.2. The second closure frontier before engineering

The manuscript then makes an important distinction. There is theorem closure, and then there is pre-engineering closure. The latter requires more than existence of admissible soft and hard sectors. It requires quantitative calibration, control closure, resource closure, material embedding, and repeatability. The integrated architecture sits exactly at the interface between theorem closure and pre-engineering closure: it uses theorem-closed route packages, but it also prepares the first composed operational target that later engineering prototypes can inherit.

5.1.3 X.45.3. Why integration is its own layer

The manuscript therefore treats integration as a genuine architectural layer. It is not merely "soft theorem plus hard theorem." It is a composed record in which soft readout prepares a later hard initialization. That already requires a shared mission object and a timing-compatibility condition. So the integrated chapter must be more than a summary of the route chapters. It must carry the first theorem-level composition law.

5.2 X.56.14. The Minimal Soft-Hard Composition Law

5.2.1 X.56.14.1. Why soft-hard composition matters

The first source-order integrated statement appears in the repeatability-and-composition block. A minimal soft-hard composition means that a soft protocol first performs remote scene acquisition or state classification and a hard protocol then uses that exported record to choose a carrier entry window and a return target. This is the first clean bridge from "seeing remotely" to "acting remotely."

The manuscript writes the composition rule through two inequalities. First, the soft record must persist long enough for hard preparation,

$$\tau_{\mathrm{mem}}^{\mathrm{soft}} > \tau_{\mathrm{prep}}^{\mathrm{hard}}.$$

Second, the uncertainty exported by the soft side must remain below the admissible hard initialization tolerance,

$$\eta_{\mathrm{soft}\to\mathrm{hard}} < \text{hard control tolerance}.$$

These inequalities are not decorative. They are the first theorem-facing form of the integrated architecture.

5.2.2 X.56.14.2. The architectural reading of the composition law

Read in the order of the manuscript, the minimal integrated architecture already has the shape

$$\text{soft scan} \longrightarrow \text{persistent decision record} \longrightarrow \text{hard entry selection} \longrightarrow \text{hard carrier cycle}.$$

The manuscript emphasizes that this order is asymmetric. Soft comes first because it prepares information. Hard comes second because it consumes that information as a control input. This asymmetry is the defining feature of the integrated layer.

5.3 X.57.C. Soft-Hard Integrated Architecture

5.3.1 X.57.13. Why one should integrate soft and hard at all

When the manuscript turns from theorem closure to architecture, it immediately notes that real architectures will likely not keep soft and hard completely isolated. A natural use case is exactly the composition law above: soft remote acquisition or classification followed by hard carrier initialization, target selection, verification targeting, or return targeting.

The architecture chapter therefore does not begin with the most complicated coupled machine. It begins with the smallest integrated use-case that respects the already constructed difference between the branches.

5.3.2 X.57.14. Integrated architecture I0

The first integrated target is $I0$.

Definition 5.1 Integrated prototype goal $I0$. A soft binary or small-alphabet scan configures a subsequent hard one-pole cycle. In source shorthand,

$$\text{Soft Scan} \to \text{Decision Record} \to \text{Hard Entry Schedule}.$$

The manuscript then lists the needed modules explicitly:

1. one soft memory-export channel,
2. one hard initialization controller,
3. one compatibility interface satisfying

$$\tau_{\text{mem}}^{\text{soft}} > \tau_{\text{prep}}^{\text{hard}}.$$

This is the first integrated architecture because it is much simpler than trying to combine full hierarchical soft-scene reconstruction with full composite hard transport immediately.

5.3.3 Mission records and readiness data

The integrated layer is organized by a mission-readiness predicate on observer states and a soft-hard integration record. Readiness is the first observer-level threshold for saying that the soft export, hard initialization, and their timing compatibility are present together in one coupled record.

Thus one begins from a readiness predicate

$$\text{ready} : \sigma \to \text{Prop},$$

and a record containing readiness data, a soft sector, and a hard carrier cycle. Mission admissibility is then

$$\text{MissionAdmissible(record)} \iff \exists s \in \sigma \text{ such that the record is ready at } s.$$

The first integrated witness is a record together with such an observer state. This is exactly the point at which integrated admissibility becomes constructive rather than only existential.

Lemma 5.2 [P]. Full FTL background yields mission admissibility together with the target suite. *If a soft-hard integration record is ready in some observer state and the full FTL background holds, then the record is mission-admissible and the full FTL target suite is available at the same time.*

The manuscript meaning of this lemma is precise. Integration does not destroy the route packages. Once the combined record is ready, the soft witness package, the hard admissibility package, and the mission package coexist in one and the same theorem layer.

Proposition 5.3 [P]. Integrated local control preserves mission admissibility. *For an integrated mission witness, local soft branch control at an active sector state yields both branch readout and mission admissibility, and under the full background this local control coexists with the full FTL target suite.*

This proposition is the theorem-level expression of the manuscript claim that soft information is not merely a preprocessing artifact. It remains genuinely active at the mission level.

5.3.4 Why $I0$ is the correct first integration

The manuscript gives three reasons.

First, binary or small-alphabet soft scan is already much simpler than full hierarchical remote-scene reconstruction.

Second, one-pole hard transport is already much simpler than composite bound transport.

Third, integrated use should begin only after each side works separately.

Thus $I0$ is the first correct integrated target because it composes the smallest stable soft block with the smallest stable hard block. It is the first bridge from route closure to architecture, not a premature total system.

5.4 X.57.D–X.57.E. Transition to the Pre-Engineering Layer

The integrated architecture chapter stops at the point where theorem-level composition has been stated cleanly. Engineering build order, notebook structure, calibration pipelines, and prototype selection belong to the pre-engineering companion material and are not part of the core theorem line of the present manuscript.

6 Proof Companion: Dependency-Complete Proofs and Proof Sketches

This chapter gives dependency-complete proofs and proof sketches for the theorem-level statements isolated in the main text. Each proof is written modulo the imported assumptions fixed in Section 2, and each argument uses only the local objects and hypotheses named in the corresponding statement.

Proofs for explicit-family results are given where the argument is directly tied to the displayed model. Proofs for the general hard closure chain H-C1 through H-C5 are recorded here in the same theorem format, using the perturbation norm, positivity margins, openness lemmas, continuity estimates, and projection-compatibility hypotheses fixed in the hard chapter. To keep the chapter uniform, every proof is organized as assumptions used, fixed objects when needed, claim, and argument.

Proof of Theorem 3.1. **Assumptions used.** Fix the four candidate classes introduced in the soft route: shared-memory bridge, branch-locked composite sector, topological soft link, and chemically information-bearing link.

Claim. Only the branch-locked composite sector can serve as the first main-line soft theorem package.

Argument. The shared-memory bridge is too weak to count as the first main-line theorem package because the remote information is already mediated by an intermediate memory structure

rather than by a direct non-factorizable witness. The topological and chemical candidates are too late because they require the many-body and materials layer that the first soft closure stage does not assume. The branch-locked composite sector occupies the only admissible middle position. It remains inside the reduced-state observer language, so it is early enough. It yields a nontrivial reduced detector shadow, so it is operationally visible. And it is not reducible to a mere pre-written memory record, so it still realizes the required non-factorizable witness logic. Therefore it is the earliest candidate satisfying all soft minimal criteria at once. □

Proof of Proposition 3.7. **Assumptions used.** Assume the soft background hypotheses stated in the theorem. By construction, that background is the conjunction of the branch-locked binary witness family and the soft repeatability family.

Claim. The bundled soft background yields exactly the minimal soft theorem package announced in the theorem.

Argument. The binary witness theorem gives positive detector contrast in the first weight-soft specialization, while the later scanner, persistence, and robustness blocks show that the same witness logic survives admissible repeated cycles strongly enough to count as an operational regime rather than as a single favorable event. One part gives readability; the other part gives persistence of that readability. Conjoining the two recovered statements yields the minimal soft theorem package. Nothing from the hard branch or mission stack is required, because the theorem is deliberately local to the soft background. □

Proof of Proposition 3.8. **Assumptions used.** Use the explicit scanner ladder X.18 through X.22 from the soft route.

Claim. Each later scanner stage contains the earlier stage as an embedded subproblem, so the ladder is cumulative.

Argument. The binary witness stage consists of two labels and a nonzero contrast between their detector shadows. The finite scanner stage replaces those two labels by an alphabet of size N, but the binary stage remains embedded as the special case obtained by restricting to any two labels b and c. Indeed, the finite scanner contrast formula

$$\Delta_{b,c}^{(N)} = \frac{4\varepsilon}{(1+\varepsilon)^2 + (N-1)(1-\varepsilon)^2}$$

reduces the operational question to the same detector distinguishability logic used in the binary case.

The hierarchical scanner stage again adds structure rather than replacing it. A hierarchical label

$$\mathbf{b} = (b^{(1)}, \ldots, b^{(L)})$$

is simply an ordered stack of finite labels, one per scale. Readability at the hierarchical level is defined by readability at each scale together with scale consistency. Therefore each scale contains a finite scanner as a subproblem, and each finite scanner contains binary witness questions as subproblems.

Because the operational content of the earlier stage survives inside the later stage, the ladder is cumulative. The later milestone does not abolish the earlier witness condition; it inherits it and adds new organization, namely a larger alphabet or a multiscale ordering. □

Proof of Theorem 4.2. **Assumptions used.** Fix the four hard carrier classes introduced in the hard route.

Claim. The branch-transfer carrier is the first hard candidate satisfying the full theorem-level requirements.

Argument. Candidate I, the desynchronized transport sector, already suggests the possibility of $c_T > c_L$, but it gives only a fast sector and not yet a stable transport cycle. Candidate III, the paired or many-body carrier, is too late because it presupposes the many-body and composite layer. Candidate IV, the projection-shortcut pseudo-carrier, is a control class and therefore cannot be the first main line.

Candidate II, the branch-transfer carrier, is the first class that carries the full cycle logic. It rewrites the hard problem in terms of the avoided-crossing family

$$\mathcal{D}_\lambda(\omega, q) = \begin{pmatrix} \omega - E_L(q) & -\lambda\Delta(q) \\ -\lambda\overline{\Delta(q)} & \omega - E_T(q) \end{pmatrix},$$

which already contains entry, propagation, and return in one explicit spectral family. That is why it is the first candidate capable of supporting one-pole transport and, later, projected-pole and identity analysis.

Therefore the branch-transfer carrier is the earliest candidate satisfying the full theorem-level requirements. It combines explicit superluminal branch selection, controlled transfer, projected-pole compatibility, and cycle-identity compatibility. □

Proof of Proposition 4.3. **Assumptions used.** Assume the hard background named in the theorem.

Claim. The hard background yields exactly the minimal hard theorem package stated in the theorem.

Argument. The theorem follows by direct unpacking of the hard background. In the normalized package,

$$\text{hard cycle background} = \text{one-pole hard-cycle admissibility} \wedge \text{composite hard-cycle repeatability}.$$

The hard background therefore recovers these two components separately: one-pole hard-cycle admissibility and composite hard-cycle repeatability.

The one-pole part corresponds to the explicit branch-transfer carrier, projected-pole survival, and the first identity-preserving cycle theorem. The composite part corresponds to carrier-phase binding and the repeatable bound-sector package. The hard background was assembled precisely so that both layers are stored in one route-level bundle.

Therefore, once the hard background is assumed, one recovers both the one-pole and composite theorem families. This is exactly the minimal hard package stated in the theorem. □

Proof of Proposition 4.11. **Assumptions used.** Compare the control quantities required at successive hard stages.

Claim. The later hard stages preserve the earlier one-pole control package while adding new obligations, so the hard ladder is cumulative.

Argument. The one-pole hard cycle already requires a nonempty superluminal window, adiabatic accessibility, projected-pole survival, and small one-pole identity loss. In the notation of the hard route, this means one must simultaneously control the gap, the adiabatic score, the residue data, and the observable drift.

The composite stage adds new control variables rather than replacing the old ones. The carrier-phase binding margin

$$B_T(q, \lambda) = E_{\mathrm{sep}}^{(T)}(q, \lambda) - E_{\mathfrak{P},*}(q, \lambda)$$

must become positive, the composite reduced resolvent must retain a pole or narrow resonance, and the composite identity loss must remain small. None of these additions remove the original one-pole obligations for the constituent sectors.

Hence the later stage is admissible only if it preserves the earlier one-pole control package while extending it by binding and composite observability. □

Proof of Lemma 5.2. **Assumptions used.** Take a soft-hard integration record with readiness data, a soft sector, and a hard cycle, and assume the record is ready in some observer state.

Claim. Mission admissibility and the full route-level target suite coexist for a ready integration record under the full FTL background.

Argument. If the record is ready in some observer state, then by definition the record is mission-admissible. This gives the first part of the conclusion.

Now assume the full FTL background. Then the route-level target suite follows: branch-locked binary witness, soft repeatability, one-pole hard-cycle admissibility, composite hard-cycle repeatability, and integrated soft-to-hard mission admissibility. The mission theorem packages these two outputs together for a ready integration record.

At theorem level this means that integration does not destroy the route packages. The record is simultaneously admissible as a mission object and still carries the entire target suite inherited from the underlying background. □

Proof of Proposition 5.3. **Assumptions used.** Take an integrated mission witness containing a mission execution witness, a branch-locked soft witness, and an alignment identity saying that the soft witness sector is the same soft sector that appears inside the mission record.

Claim. Local soft branch control remains available inside the integrated mission record, and under the full background it coexists with mission admissibility and the full target suite.

Argument. Suppose a sector state is active for the mission record. By the alignment identity, the same state is active for the branch witness. Therefore the branch witness applies and yields branch readout at that state.

At the same time, the mission execution witness already carries mission admissibility, because it consists of a record together with an observer state at which readiness is realized. Hence local branch readout and mission admissibility are available simultaneously. This is exactly the local content of the integrated architecture: the soft side remains operationally active after composition rather than disappearing into a background preprocessing step.

If the full background is additionally assumed, the route-level target suite is recovered as in the previous theorem. Therefore one may state all three facts together: local soft branch control, mission admissibility of the coupled record, and coexistence with the full target suite. □

Additional Hard-Route Proofs

Proof sketch for X.26.5 (Hard observability theorem). **Assumptions used.** Assume a branch-transfer family with a selected branch, a nonempty superluminal window, small adiabatic score, small cycle loss, and a projected physical resolvent block of the form

$$P_{\sigma_{\mathfrak{p}}} \mathcal{R}_{\mathrm{phys}} P_{\sigma_{\mathfrak{p}}} \sim \frac{Z_*(q,\lambda)}{\eta - E_*(q,\lambda)}$$

with $Z_* > 0$.

Fixed objects. Fix the selected transported branch E_*, its residue Z_*, and the observer projector selecting the physical one-pole sector.

Claim. The family realizes a physically admissible one-pole hard carrier.

Argument. The projected physical resolvent already contains the required simple pole with positive residue, so the transported sector is physically observable rather than merely internal. The superluminal window supplies the observer-relative hard-transport inequality. Small adiabatic score keeps the transport cycle on the selected branch, and small cycle loss prevents the object-sector from being destroyed during entry and return. These conditions are exactly the route-level observability package named in X.26, so their conjunction yields a physically admissible one-pole hard carrier. □

Proof sketch for X.27.9 and X.27.11 (Pole survival theorem and physical pole corollary). **Assumptions used.** Use the explicit two-branch hard carrier family of X.25 together with $\lambda > 0$, a positive avoided-crossing gap, and a nonvanishing residue numerator on the transport window.

Fixed objects. Fix the determinant factorization

$$\det \mathcal{D}_\lambda(\omega, q) = (\omega - E_+(q,\lambda))(\omega - E_-(q,\lambda))$$

and the projected luminal-sector resolvent R_L.

Claim. The selected branch remains a simple nonzero-residue projected pole, and if the transported sector has nonzero overlap with P_{phys} while the projector class stays fixed, then this pole is also a physical hard pole.

Argument. The determinant factorization shows that the poles are exactly E_+ and E_-, while the positive gap makes them distinct and therefore simple. The residue formulas in X.27.6 are finite and nonzero precisely when the corresponding numerator does not vanish, so the selected transported branch remains a simple nonzero-residue projected pole throughout the working window. Once non-annihilation by P_{phys} and projector-class stability are assumed, the projected pole survives under the physical projector as stated in X.27.10. That gives both the sector-resolved pole-survival theorem and the physical pole corollary. □

Proof sketch for X.28.8 (Identity-preserving hard cycle). **Assumptions used.** Assume sector-resolved pole survival on the selected branch, non-annihilation by P_{phys}, projector-class stability on the working window, adiabatic accessibility, small residue drift, and small conserved-observable drift.

Fixed objects. Fix the cycle-identity functional

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} = w_{\mathrm{st}}\epsilon_{\mathrm{st}} + w_Z\epsilon_Z + w_Q\epsilon_Q.$$

Claim. The full cycle is identity-preserving at the first observer-relevant hard level.

Argument. X.28.6 already gives the control estimate

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} \lesssim w_{\mathrm{st}}L_{\mathrm{nad}} + w_Z\epsilon_Z + w_Q\epsilon_Q.$$

Adiabatic accessibility makes L_{nad} small, while the explicit hypotheses make ϵ_Z and ϵ_Q small. Pole survival prevents the cycle from losing the transported one-pole sector in the carrier phase. Therefore the identity functional stays small, which is exactly the identity-preserving hard-cycle claim of X.28.8. □

Proof sketch for X.29.7 (The first composite hard theorem). **Assumptions used.** Assume that each constituent one-pole sector satisfies the one-pole hard theorem, that a composite carrier branch exists, that the carrier-phase binding margin is positive on the transport window, that the composite reduced resolvent has an isolated pole or narrow resonance, and that the composite cycle identity loss is small.

Fixed objects. Fix the carrier-phase binding margin

$$B_T(q,\lambda) = E^{(T)}_{\mathrm{sep}}(q,\lambda) - E_{\mathfrak{P},*}(q,\lambda)$$

and the composite projected pole data.

Claim. The composite object realizes a composite hard transport phase.

Argument. The constituent one-pole theorems supply admissible carrier transport for each component. Positive binding margin prevents breakup inside the carrier phase. The isolated composite pole or narrow resonance provides physical composite observability. Small composite identity loss ensures that the cycle returns the same bound object rather than only a fast aggregate with altered structure. Finally, the composite branch is superluminal on a nonempty window by hypothesis. These are precisely the conditions listed in X.29.6, so the composite hard theorem follows. □

Proof sketch for X.31.8 (The hard robustness theorem). **Assumptions used.** Assume the explicit hard package has a one-pole branch-transfer carrier family, projected simple-pole survival, small one-pole cycle identity loss, an explicit composite carrier model with positive binding margin and composite pole, and robustness loads below the corresponding positive margins.

Claim. Hard transport survives admissible perturbation as a robust theorem package at both the one-pole and composite levels.

Argument. The one-pole margins protect speed, gap, residue, and identity. If the one-pole perturbation load stays below their minimum, then continuity preserves the nonempty superluminal window, simple nonzero projected pole, and small one-pole identity loss. The composite margins similarly protect binding, composite pole weight, and composite identity. If the composite perturbation load stays below their minimum, then continuity preserves positive carrier-phase binding and composite observability. Since the explicit hard package is the conjunction of these one-pole and composite structures, preserving both packages preserves the full hard theorem package under admissible perturbations. □

Proofs for the General Hard Closure Chain

Proof of Lemma 4.6. **Assumptions used.** Fix the explicit branch-transfer family and a positive velocity margin $M_{\mathrm{vel}} > 0$ on U_T, within the admissible hard perturbation class of Definition 4.5.

Claim. A sufficiently small admissible perturbation preserves a nonempty superluminal carrier window.

Argument. On the explicit family one has

$$\inf_{q\in U_T} \left(|\partial_q E_T(q)| - c_L\right) = M_{\mathrm{vel}} > 0.$$

Because the hard perturbation norm controls both the branch kernel and the switching profile, the perturbed branch slope varies continuously with $\|\delta\|_{\mathrm{hard}}$. Hence there exists $C_{\mathrm{vel}} > 0$ such that

$$\sup_{q\in U_T} |\partial_q E^{\delta}_T(q) - \partial_q E_T(q)| \le C_{\mathrm{vel}}\|\delta\|_{\mathrm{hard}}.$$

Choose $\varepsilon_{\mathrm{vel}} < M_{\mathrm{vel}}/C_{\mathrm{vel}}$. Then every perturbation with $\|\delta\|_{\mathrm{hard}} < \varepsilon_{\mathrm{vel}}$ satisfies

$$|\partial_q E^{\delta}_T(q)| - c_L \ge M_{\mathrm{vel}} - C_{\mathrm{vel}}\|\delta\|_{\mathrm{hard}} > 0$$

on a nonempty subwindow of U_T. This proves openness of the nonempty carrier window. □

Proof of Lemma 4.7. **Assumptions used.** Fix positive pole-isolation and residue margins $M_{\mathrm{gap}} > 0$ and $M_Z > 0$ on the working window, within the admissible hard perturbation class of Definition 4.5.

Claim. Simplicity and nonzero residue persist under sufficiently small admissible perturbations.

Argument. Pole simplicity is protected by the gap bound

$$\Delta_{\mathrm{gap}}(q, \lambda) \geq M_{\mathrm{gap}} > 0,$$

while residue nonvanishing is protected by

$$|Z_*(q, \lambda)| \geq M_Z > 0.$$

The resolvent denominator and residue numerator depend continuously on the perturbation data entering $\|\delta\|_{\mathrm{hard}}$. Therefore there is a constant $C_{\mathrm{pole}} > 0$ such that both the gap and residue vary by at most $C_{\mathrm{pole}}\|\delta\|_{\mathrm{hard}}$. Choosing

$$\varepsilon_{\mathrm{pole}} < \frac{1}{C_{\mathrm{pole}}} \min\{M_{\mathrm{gap}}, M_Z\}$$

keeps the perturbed gap and perturbed residue strictly positive. The selected pole therefore remains simple and nonzero-residue. □

Proof of Lemma 4.8. **Assumptions used.** Fix a one-pole cycle with positive identity margin $M_{\mathrm{id}} > 0$ and the identity functional of X.28.4, within the admissible hard perturbation class of Definition 4.5.

Claim. The one-pole identity functional is continuous in the hard perturbation norm.

Argument. The functional

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} = w_{\mathrm{st}}\epsilon_{\mathrm{st}} + w_Z\epsilon_Z + w_Q\epsilon_Q$$

is a finite weighted sum of branch overlap, residue drift, and observable drift. Each term is continuous in the perturbed branch kernel, projector, and switching profile, all of which are controlled by $\|\delta\|_{\mathrm{hard}}$. Hence there exists $C_{\mathrm{id}} > 0$ with

$$|\epsilon_{\mathrm{id}}^{\mathrm{cyc},\delta} - \epsilon_{\mathrm{id}}^{\mathrm{cyc}}| \leq C_{\mathrm{id}}\|\delta\|_{\mathrm{hard}}.$$

If $C_{\mathrm{id}}\|\delta\|_{\mathrm{hard}} < M_{\mathrm{id}}$, then the perturbed cycle remains below the identity threshold and the identity regime persists. □

Proof of Lemma 4.9. **Assumptions used.** Fix positive binding, composite-residue, narrow-resonance, and composite-identity margins on the explicit composite branch, within the admissible hard perturbation class of Definition 4.5.

Claim. Positive carrier-phase binding, composite observability, and composite identity compatibility persist under sufficiently small admissible perturbations.

Argument. The binding margin, composite residue, composite width, and composite identity functional are continuous in the perturbed composite kernel. Thus there exist constants $C_{\mathrm{comp}} > 0$ and $C_{\mathrm{compid}} > 0$ such that

$$|B_T^\delta - B_T| + |Z_{\mathfrak{P}}^\delta - Z_{\mathfrak{P}}| + |\Gamma_{\mathfrak{P}}^\delta - \Gamma_{\mathfrak{P}}| \leq C_{\mathrm{comp}}\|\delta\|_{\mathrm{hard}}.$$

and

$$|\epsilon_{\mathrm{id}}^{\mathrm{comp},\delta} - \epsilon_{\mathrm{id}}^{\mathrm{comp}}| \leq C_{\mathrm{compid}}\|\delta\|_{\mathrm{hard}}.$$

Choose

$$\varepsilon_{\mathrm{comp}} < \min\left\{ \frac{1}{C_{\mathrm{comp}}} \min\{M_{\mathrm{bind}}, M_{\mathrm{comp}Z}, M_\Gamma\}, \frac{M_{\mathrm{compid}}}{C_{\mathrm{compid}}} \right\}.$$

Then the perturbed composite branch keeps positive binding margin, nonzero composite residue, and either zero width or width still below the narrow-resonance threshold 1/10. The same bound yields

$$\epsilon_{\mathrm{id}}^{\mathrm{comp},\delta} < 1.$$

Hence composite observability, binding, and identity compatibility remain valid. □

Proof of Lemma 4.10. **Assumptions used.** Fix a positive projector overlap margin $\kappa_{\mathrm{phys}} > 0$, pole isolation on the working window, and projector-class stability, within the admissible hard perturbation class of Definition 4.5.

Claim. Physical non-annihilation and projection-class stability persist under sufficiently small admissible perturbations.

Argument. The operator norm bound on δP_{phys} implies continuity of the overlap operator $P_{\mathrm{phys}}\Pi_*$. Hence there exists $C_{\mathrm{phys}} > 0$ such that

$$\|P_{\mathrm{phys}}^\delta \Pi_*^\delta - P_{\mathrm{phys}}\Pi_*\|_{\mathrm{op}} \le C_{\mathrm{phys}} \|\delta\|_{\mathrm{hard}}.$$

Choosing $\varepsilon_{\mathrm{phys}} < \kappa_{\mathrm{phys}}/C_{\mathrm{phys}}$ preserves a strictly positive overlap and therefore prevents annihilation by the physical projector. By hypothesis the perturbation class does not leave the fixed superselection sector, so the projector class remains unchanged as well. □

Proof of H-C1 (General carrier existence). **Assumptions used.** Use the explicit branch-transfer hard family of X.25, the positive velocity margin $M_{\mathrm{vel}} > 0$, and Lemma 4.6, within the admissible hard perturbation class of Definition 4.5.

Claim. There exists a nontrivial admissible class of hard carrier sectors with nonempty superluminal transport window.

Argument. The explicit family already exhibits a strict superluminal margin. Lemma 4.6 upgrades that local fact to an open perturbative statement: for all admissible deformations inside a fixed $\varepsilon_{\mathrm{vel}}$-ball in $\|\delta\|_{\mathrm{hard}}$, one retains a nonempty transport subwindow with $|\partial_q E_T^\delta(q)| > c_L$. Therefore the explicit branch-transfer family extends to a nontrivial admissible class of hard carriers, which is exactly H-C1. □

Proof of H-C2 (General projected-physical observability). **Assumptions used.** Use H-C1, sector-resolved pole survival, positive pole-isolation and residue margins, positive projector overlap margin $\kappa_{\mathrm{phys}} > 0$, and Lemmas 4.7 and 4.10, within the admissible hard perturbation class of Definition 4.5.

Claim. The selected transported branch survives as a physical asymptotic pole of the selected object-sector block throughout the admissible class.

Argument. Lemma 4.7 preserves the sector-resolved simple pole and its nonzero residue. Lemma 4.10 preserves non-annihilation by P_{phys} and keeps the physical projector inside the same class. Hence the selected transported sector continues to contribute to the projected physical resolvent, and the pole-isolation hypothesis prevents the projected block from mixing it into a degenerate continuum. Therefore

$$P_{\sigma_p} P_{\mathrm{phys}} \mathcal{R}_{\mathrm{micro}} P_{\mathrm{phys}} P_{\sigma_p} \sim \frac{Z_*^\delta(q)}{\eta - E_*^\delta(q)}$$

with $Z_*^\delta(q) \neq 0$ on the admissible neighborhood. This is H-C2. □

Proof of H-C3 (General identity preservation). **Assumptions used.** Use H-C2, the identity-control estimate of X.28, a positive identity margin $M_{\mathrm{id}} > 0$, and Lemma 4.8, within the admissible hard perturbation class of Definition 4.5.

Claim. There exists a switching regime in which the full cycle preserves the one-pole object-sector throughout the admissible class.

Argument. The explicit cycle satisfies the identity estimate of X.28.6 and therefore starts with a strictly positive margin below the admissible threshold. Lemma 4.8 shows that the full identity functional moves by at most $C_{\mathrm{id}}\|\delta\|_{\mathrm{hard}}$. Therefore every admissible perturbation with $C_{\mathrm{id}}\|\delta\|_{\mathrm{hard}} < M_{\mathrm{id}}$ preserves the one-pole identity regime. Together with H-C2, this keeps both physical observability and object identity in the same perturbative neighborhood, proving H-C3. □

Proof of H-C4 (General composite observability, binding, and identity compatibility). **Assumptions used.** Use H-C3, the explicit composite theorem of X.29, positive composite margins M_{bind}, $M_{\mathrm{comp}Z}$, M_{Γ}, a positive composite identity margin $M_{\mathrm{compid}} > 0$, and Lemma 4.9, within the admissible hard perturbation class of Definition 4.5.

Claim. There exists an admissible class of composite carrier sectors with positive binding margin, physical composite observability, and composite identity compatibility.

Argument. The explicit composite prototype has positive binding margin, nonzero residue, and either zero width or width below the narrow-resonance threshold. Lemma 4.9 preserves not only positive carrier-phase binding and composite observability, but also composite identity compatibility, because

$$|\epsilon_{\mathrm{id}}^{\mathrm{comp},\delta} - \epsilon_{\mathrm{id}}^{\mathrm{comp}}| \le C_{\mathrm{compid}}\|\delta\|_{\mathrm{hard}} < M_{\mathrm{compid}}.$$

Therefore

$$\epsilon_{\mathrm{id}}^{\mathrm{comp},\delta} < 1.$$

H-C3 simultaneously preserves the one-pole constituent cycles needed to define the composite sector. Therefore the composite transported branch remains bound, observable, identity-compatible, and superluminal on that neighborhood. This is H-C4. □

Proof of H-C5 (Hard robustness lifting). **Assumptions used.** Use H-C4 together with the robustness margins and loads of X.31.4 through X.31.7, within the admissible hard perturbation class of Definition 4.5.

Claim. The entire hard theorem package is stable on an admissible perturbative neighborhood.

Argument. The one-pole inequalities

$$R_{1\mathrm{p}} < \min\{M_{\mathrm{vel}}, M_{\mathrm{gap}}, M_Z, M_{\mathrm{id}}\}$$

keep the transport window, simple projected pole, and identity control intact. The composite inequalities

$$R_{\mathrm{comp}} < \min\{M_{\mathrm{bind}}, M_{\mathrm{comp}Z}, M_{\Gamma}, M_{\mathrm{compid}}\}$$

keep positive binding, composite observability, and composite identity intact. Since H-C1 through H-C4 identify exactly the ingredients protected by these margins, the conjunction of the one-pole and composite robustness inequalities lifts the entire package to an admissible neighborhood. This is H-C5. □

A Material Assumptions and Pre-Engineering Constraints

This appendix collects the late pre-engineering block of the manuscript in companion form. The order is still the manuscript order: first the distinction between theorem closure and pre-engineering closure, then the six pre-engineering blocks E-C1 through E-C6, and then the explicit calibration, control, resource, and material-embedding chains. Its role is to preserve the theorem-to-prototype transition without interrupting the theory core established in the main text.

The manuscript makes one decisive distinction immediately. A branch may be theorem-closed and still not be prototype-ready. Between route-level theorems and actual devices there is an intermediate closure layer: quantitative calibration, control closure, resource closure, material embedding, and repeatability. The present chapter records that layer in the same direct order as the manuscript.

A.1 X.45–X.46. From Theorem Closure to Pre-Engineering Closure

A.1.1 Two closure frontiers

The manuscript says that there are now two different closure frontiers.

Level 1 is theorem closure. On that level, soft FTL is already closed as a robust hierarchical remote-scene approximation package, and hard FTL is already closed as an audited open-class hard transport package.

Level 2 is pre-engineering closure. The manuscript defines this second level by four additional requirements beyond theorem closure: quantitative calibration closure, control closure, resource closure, and material embedding closure. Repeatability and protocol composition are then added as the last pre-engineering theorem block before prototype design.

In source form, the remaining pre-engineering blocks are listed in the exact order

$$E\text{-}C1 \to E\text{-}C2 \to E\text{-}C3 \to E\text{-}C4 \to E\text{-}C5 \to E\text{-}C6.$$

The chapter follows the same order.

A.1.2 What pre-engineering closure means

The manuscript gives the fully explicit definition. Full theoretical closure before engineering means the conjunction of:

1. theorem-package closure,
2. quantitative calibration closure,
3. control closure,
4. resource closure,
5. material embedding closure,
6. repeatability closure.

Only after these six layers are closed may one pass from pure theory to engineering prototypes without a methodological gap.

A.2 X.51. E-C1: Quantitative Calibration for Soft FTL

A.2.1 Why soft calibration comes first

The manuscript begins the pre-engineering chain with soft calibration because the soft branch is already the more mature route. The question is no longer whether a hierarchical scanner package exists. The question is how to compute, from soft kernel data, the readout margins, times, capacities, and hierarchy depth.

The design question is therefore

$$\boxed{\text{how does one compute } \Delta_{\min},\ t_{\mathrm{read}},\ \eta_{\mathrm{mem}},\ L_{\max},\ N_{\max} \text{ from soft-sector data?}}$$

A.2.2 Input data and calibration observables

At scale ℓ, the manuscript takes as input branch amplitudes

$$a_k^{(\ell,b)}, \qquad k = 1, \dots, N_\ell, \qquad b \in \Sigma^{(\ell)},$$

from which normalized detector probabilities are formed,

$$p_k^{(\ell,b)} = \frac{|a_k^{(\ell,b)}|^2}{\sum_j |a_j^{(\ell,b)}|^2}.$$

The manuscript then reads the first calibration observable as pairwise scanner contrast,

$$\Delta_{b,c}^{(\ell)} := \frac{1}{2} \sum_{k=1}^{N_\ell} |p_k^{(\ell,b)} - p_k^{(\ell,c)}|, \qquad \Delta_{\min}^{(\ell)} := \min_{b \neq c} \Delta_{b,c}^{(\ell)}.$$

For the weight-soft specialization this becomes

$$\Delta_{\min}^{(\ell)} = \frac{4\varepsilon_\ell}{(1+\varepsilon_\ell)^2 + (N_\ell - 1)(1-\varepsilon_\ell)^2}.$$

The second observable is the minimal readout time. Using the classicalization law

$$\epsilon_{\mathrm{cl}}^{(\ell)}(t) \leq C_\ell e^{-\Gamma_{\mathrm{dec},\ell} t},$$

and a detector error target $\eta_{\mathrm{cl}}^{(\ell)}$, the manuscript obtains the design law

$$\boxed{t_{\mathrm{read}}^{(\ell)} \geq \frac{1}{\Gamma_{\mathrm{dec},\ell}} \log \frac{C_\ell}{\eta_{\mathrm{cl}}^{(\ell)}}.}$$

The third observable is the memory-write distortion,

$$\eta_{\mathrm{mem}}^{(\ell)} := \sup_b \left| \rho_{\mathrm{mem}}^{(\ell,b)} - |m_b^{(\ell)}\rangle\langle m_b^{(\ell)}| \right|_1,$$

with the necessary condition that observer information gain is positive and

$$\eta_{\mathrm{mem}}^{(\ell)} < \Delta_{\min}^{(\ell)}.$$

The fourth observable is the statistical readout budget $\eta_{\mathrm{stat}}^{(\ell)}$, scaling like $O(n_\ell^{-1/2})$ for n_ℓ local samples. The manuscript then combines all soft operational errors into

$$R_{\mathrm{soft}}^{(\ell)} := \eta_{\mathrm{cl}}^{(\ell)} + \eta_{\mathrm{mem}}^{(\ell)} + \eta_{\mathrm{stat}}^{(\ell)},$$

and the operational readability condition becomes

$$\boxed{R_{\mathrm{soft}}^{(\ell)} < \Delta_{\min}^{(\ell)}.}$$

A.2.3 Capacity and hierarchy laws

Solving the weight-soft contrast inequality for the alphabet size gives the first explicit alphabet-capacity law,

$$\boxed{N_\ell < 1 + \frac{\frac{4\varepsilon_\ell}{R_{\mathrm{soft}}^{(\ell)}} - (1+\varepsilon_\ell)^2}{(1-\varepsilon_\ell)^2}.}$$

The hierarchy depth is then determined by the largest L such that every active scale satisfies the readability inequality,

$$L_{\max} := \max\left\{L:\ R_{\mathrm{soft}}^{(\ell)} < \Delta_{\min}^{(\ell)} \text{ for all } \ell \le L\right\}.$$

The manuscript also adds the memory-persistence condition

$$\tau_{\mathrm{mem}}^{(\ell)} > t_{\mathrm{read}}^{(\ell)} + t_{\mathrm{write}}^{(\ell)} + t_{\mathrm{dec}}^{(\ell)}.$$

A.2.4 Meaning of E-C1

The manuscript concludes E-C1 by stating that soft FTL is no longer only theorem-closed. It is now quantitatively calibratable: from soft kernel data one gets a computable recipe for contrast, readout time, alphabet capacity, and hierarchy depth.

A.3 X.52. E-C2: Quantitative Calibration for Hard FTL

A.3.1 Why hard calibration comes next

The hard branch must now be translated into the same design language. The input data are the explicit branch-transfer carrier family,

$$\mathcal{D}_\lambda(\omega, q) = \begin{pmatrix} \omega - E_L(q) & -\lambda\Delta_0 \\ -\lambda\Delta_0 & \omega - E_T(q) \end{pmatrix}, \qquad E_L(q) = \sqrt{m^2 + c_L^2 q^2}, \qquad E_T(q) = \delta + v_T|q|,$$

together with the selected branches $E_\pm(q,\lambda)$, the avoided-crossing gap $\Delta_{\mathrm{gap}}(q,\lambda)$, the one-pole identity package, and the composite carrier data.

The design question is therefore

how does one compute the admissible hard transport region from carrier-family data?

A.3.2 Hard calibration observables

The first hard observable is the superluminal carrier window

$$U_T := \{q:\ |\partial_q E_*(q,1)| > c_L\},$$

together with the positive carrier margin

$$M_{\mathrm{vel}} := \inf_{q\in U_T} (|\partial_q E_*(q,1)| - c_L).$$

The second observable is the switching-time law. Imposing an adiabaticity budget

$$\mathcal{A}(q,t) = \frac{|\dot\lambda(t)|\Delta_0}{\Delta_{\mathrm{gap}}(q,\lambda)^2} \le \alpha_{\mathrm{ad}}, \qquad \alpha_{\mathrm{ad}} \ll 1,$$

yields the lower bound

$$\boxed{\tau_{\mathrm{sw}}^{\min} \geq \frac{\Delta_0}{\alpha_{\mathrm{ad}}(\Delta_{\mathrm{gap}}^{\min})^2}.}$$

The third observable is the pole-survival margin,

$$M_Z := \inf_{(q,\lambda)\in W_{\mathrm{tr}}} |Z_*(q,\lambda)|.$$

The fourth observable is the one-pole identity budget,

$$\epsilon_{\mathrm{id}}^{\mathrm{1p}} = w_E\epsilon_E + w_Z\epsilon_Z + w_Q\epsilon_Q + w_\chi\epsilon_\chi,$$

controlled by adiabatic leakage plus residue, observable, and observer-shadow drift.

The fifth observable is the composite binding margin,

$$M_{\mathrm{bind}} := \inf_{(Q,\lambda)\in W_{\mathfrak{P}}} B_T(Q,\lambda),$$

which reduces to B_0 in the simplest explicit bound-transport prototype.

The sixth observable is the composite identity budget,

$$\epsilon_{\mathrm{id}}^{\mathrm{comp}} = w_{\mathrm{pole}}\epsilon_{\mathrm{pole}}^{\mathrm{comp}} + w_Z\epsilon_Z^{\mathrm{comp}} + w_Q\epsilon_Q^{\mathrm{comp}} + w_B\epsilon_B^{\mathrm{comp}}.$$

These are assembled into the hard calibration region $\mathcal{R}_{\mathrm{hard}}^{\mathrm{cal}}$ by the simultaneous conditions

$$M_{\mathrm{vel}} > 0, \qquad \tau_{\mathrm{sw}} \geq \tau_{\mathrm{sw}}^{\min}, \qquad M_Z > 0, \qquad \epsilon_{\mathrm{id}}^{\mathrm{1p}} \leq \eta_{\mathrm{1p}}, \qquad M_{\mathrm{bind}} > 0, \qquad \epsilon_{\mathrm{id}}^{\mathrm{comp}} \leq \eta_{\mathrm{comp}}.$$

A.3.3 Meaning of E-C2

The manuscript concludes that hard FTL is now quantitatively calibratable in the same design-oriented sense as soft FTL. Given hard carrier data, one now has a computable recipe for switching time, carrier margin, pole margin, and composite feasibility.

A.4 X.53. E-C3: Control-Theoretic Closure

A.4.1 Why calibration is not enough

After E-C1 and E-C2, the manuscript insists that calibration alone does not produce a usable protocol. One must convert a feasible region into an explicit control family. Control closure therefore means the existence of piecewise controls that initialize the protocol, keep it inside the admissible window, terminate it cleanly, and make repetition possible.

A.4.2 Soft control state and cycle

For soft FTL the observer-level control state is

$$X_{\mathrm{soft}}(t) = \big(\rho_D(t), \rho_{\mathrm{mem}}(t), h(t)\big),$$

with control input

$$u_{\mathrm{soft}}(t) = \big(g_{\mathrm{link}}(t), g_{\mathrm{det}}(t), g_{\mathrm{write}}(t), g_{\mathrm{reset}}(t), h_{\mathrm{cmd}}(t)\big).$$

The manuscript then writes the soft control cycle in four phases: initialization, link-and-read, write-and-freeze, and decode-and-reset. A soft stabilizing control family is one that keeps

$$R_{\mathrm{soft}}^{(\ell)}(t) < \Delta_{\mathrm{min}}^{(\ell)}$$

throughout the read/write interval and preserves the memory-persistence inequality.

A.4.3 Hard control state and cycle

For hard FTL the control state is spectral rather than informational,

$$X_{\mathrm{hard}}(t) = (\lambda(t), q(t), \mathfrak{p}(t), \mathfrak{P}(t)),$$

with control input

$$u_{\mathrm{hard}}(t) = \big(u_{\mathrm{sw}}(t), u_{\mathrm{hold}}(t), u_{\mathrm{exit}}(t), u_{\mathrm{ver}}(t)\big).$$

The cycle is again four-phase: entry, carrier hold, exit, and verification. A hard stabilizing control family is one that keeps the process inside $\mathcal{R}^{\mathrm{cal}}_{\mathrm{hard}}$, preserving velocity, pole, identity, and binding margins.

A.4.4 Meaning of E-C3

The manuscript closes E-C3 by stating that both soft and hard FTL are not only theorem-closed and calibratable, but control-closed. Soft control is information-stabilizing; hard control is identity-and-binding-stabilizing.

A.5 X.54. E-C4: Resource and Thermodynamic Closure

A.5.1 Why a resource layer is mandatory

The manuscript next asks what one cycle costs. A regime that works only as an uncontrolled energetic accident is not prototype-ready. Resource closure therefore means that each cycle has an explicit action, work, dissipation, memory, and stabilization budget together with a feasibility inequality for repeated operation.

A.5.2 Soft cycle cost

The manuscript decomposes the soft cycle cost into

$$\mathcal{C}^{\mathrm{cyc}}_{\mathrm{soft}} = W_{\mathrm{link}} + W_{\mathrm{det}} + W_{\mathrm{write}} + W_{\mathrm{erase}} + Q^{\mathrm{diss}}_{\mathrm{soft}}.$$

Combined with the already calibrated readability condition,

$$R^{(\ell)}_{\mathrm{soft}} < \Delta^{(\ell)}_{\mathrm{min}},$$

this yields the soft resource-feasibility inequality

$$\boxed{J^{\mathrm{soft}}_{\mathrm{in}} > \frac{W_{\mathrm{link}} + W_{\mathrm{det}} + W_{\mathrm{write}} + W_{\mathrm{erase}} + Q^{\mathrm{diss}}_{\mathrm{soft}}}{\tau^{\mathrm{soft}}_{\mathrm{cyc}}}.}$$

A.5.3 Hard cycle cost

The hard cycle cost is written as

$$\mathcal{C}^{\mathrm{cyc}}_{\mathrm{hard}} = W_{\mathrm{entry}} + W_{\mathrm{hold}} + W_{\mathrm{exit}} + W_{\mathrm{ver}} + W_{\mathrm{bind}} + Q^{\mathrm{diss}}_{\mathrm{hard}}.$$

This must hold together with the calibrated spectral and identity inequalities, so the hard feasibility condition becomes

$$\boxed{J^{\mathrm{hard}}_{\mathrm{in}} > \frac{W_{\mathrm{entry}} + W_{\mathrm{hold}} + W_{\mathrm{exit}} + W_{\mathrm{ver}} + W_{\mathrm{bind}} + Q^{\mathrm{diss}}_{\mathrm{hard}}}{\tau^{\mathrm{hard}}_{\mathrm{cyc}}}.}$$

The manuscript then states the common pattern explicitly: a useful cycle is admissible dynamics plus surviving invariants plus positive resource budget.

A.5.4 Meaning of E-C4

After E-C4, both branches are theorem-closed, calibratable, control-closed, and resource-closed.

A.6 X.55. E-C5: Material Embedding

A.6.1 Why medium classes are part of closure

The manuscript next asks what physical medium classes can actually realize the required sector architectures. This is not a decorative appendix. Without a nonempty medium class $\mathcal{M}$, there is no prototype, even if the sector theory is closed.

A.6.2 Soft material classes

A soft-compatible medium must support simultaneously a branch-locked or multi-window soft sector, detector classicalization, persistent memory, and multiscale coarse readout. The manuscript uses the many-body and chemistry chapters to name the candidate medium classes.

The first class is chemical-memory media satisfying the chemical memory criterion

$$\tau_{\text{flip}} \gg \tau_{\text{read}}, \qquad D_{\text{tr}}(\rho_{X_0}, \rho_{X_1}) \approx 1.$$

The second class is glassy or polymeric information-bearing media, using slow relaxation and long-chain assembly structure. The third class is topological or gapped many-body media, attractive when additional robustness against local perturbations is required.

The manuscript therefore treats the soft material target class as follows:

$\mathcal{M}_{\text{soft}}$ is built from chemical-memory media, slow metastable materials, and gapped or topological many-body phases.

A.6.3 Hard material classes

Hard-compatible media must support branch-transfer interfaces, superluminal carrier windows, physically observable one-pole carrier phases, positive binding margin for bound objects, and control interfaces for entry, hold, exit, and verification.

The manuscript again identifies three natural classes: paired or superconducting-like carrier media, topological defect or quasiparticle media, and composite chemical-material carrier media. This yields the hard material target class:

$\mathcal{M}_{\text{hard}}$ is built from paired or quasiparticle media, topological-defect media, and composite material carrier phases.

A.6.4 Meaning of E-C5

The unified material-embedding closure theorem states that if soft and hard media satisfy the corresponding medium criteria, then the FTL program is material-embedded at the pre-engineering theorem level.

A.7 X.56. E-C6: Repeatability and Protocol Composition

A.7.1 Why repeatability is the last theoretical barrier

After E-C5, the only remaining pre-engineering question is whether the protocols can be run repeatedly and composed. A one-cycle success is not yet a prototype. The manuscript therefore introduces per-cycle drift, reset or refresh conditions, and composition laws.

A.7.2 Soft repeatability

For soft FTL one writes the cycle recursion

$$X_{\mathrm{soft}}(m) = (\rho_D^{(m)}, \rho_{\mathrm{mem}}^{(m)}, h^{(m)}), \qquad \mathcal{T}_{\mathrm{soft}} : X_{\mathrm{soft}}(m) \mapsto X_{\mathrm{soft}}(m+1),$$

and defines the accumulated effective load

$$R_{\mathrm{soft,eff}}^{(\ell)}(m) = R_{\mathrm{soft}}^{(\ell)} + \delta_{\mathrm{rep}}^{(\ell)}(m).$$

Repeatability up to cycle number n requires

$$R_{\mathrm{soft,eff}}^{(\ell)}(m) < \Delta_{\mathrm{min}}^{(\ell)} \qquad \text{for all } m \le n.$$

Reset quality is controlled by detector-reset and memory-reset deviations, and hierarchical composition is then read off from scale-wise satisfaction of the same inequality. The maximal repeatable hierarchy depth becomes

$$L_{\mathrm{rep}}^{\max}(n) = \max\left\{ L : \; R_{\mathrm{soft,eff}}^{(\ell)}(m) < \Delta_{\mathrm{min}}^{(\ell)} \; \forall \ell \le L, \; \forall m \le n \right\}.$$

A.7.3 Hard repeatability

For hard FTL one instead tracks cumulative drift in the carrier margins,

$$\delta_{\mathrm{vel}}(m), \quad \delta_Z(m), \quad \delta_{\mathrm{id}}^{\mathrm{1p}}(m), \quad \delta_{\mathrm{bind}}(m), \quad \delta_{\mathrm{id}}^{\mathrm{comp}}(m),$$

and requires the effective conditions

$$M_{\mathrm{vel}} - \delta_{\mathrm{vel}}(m) > 0, \qquad M_Z - \delta_Z(m) > 0, \qquad \epsilon_{\mathrm{id}}^{\mathrm{1p}} + \delta_{\mathrm{id}}^{\mathrm{1p}}(m) \le \eta_{\mathrm{1p}},$$

$$M_{\mathrm{bind}} - \delta_{\mathrm{bind}}(m) > 0, \qquad \epsilon_{\mathrm{id}}^{\mathrm{comp}} + \delta_{\mathrm{id}}^{\mathrm{comp}}(m) \le \eta_{\mathrm{comp}}.$$

A refresh strategy must keep these cumulative drifts below the corresponding positive margins. That is the manuscript's hard n-cycle stability criterion.

A.7.4 Protocol composition

The manuscript then states three composition laws.

Soft-soft composition is feasible when the sum of effective loads remains below the corresponding contrasts.

Hard-hard composition is feasible when the refreshed output state of the first hard module lies inside the calibrated admissible region of the second module.

Soft-hard composition is feasible exactly when the soft record remains persistent and accurate over the hard initialization time, i.e.

$$\tau_{\mathrm{mem}}^{\mathrm{soft}} > \tau_{\mathrm{prep}}^{\mathrm{hard}}, \qquad \eta_{\mathrm{soft}\to\mathrm{hard}} < \text{hard control tolerance}.$$

This is again the same composition law that underlies the integrated architecture chapter.

A.7.5 Meaning of E-C6

The unified repeatability theorem says that if soft and hard FTL are material-embedded, resource-closed, and control-closed, and if reset or refresh strategies keep cumulative drifts below the calibrated margins, then both branches admit n-cycle repeatability, in-branch protocol composition, and soft-to-hard composition across branches.

At this point, the manuscript states the full pre-prototype closure conclusion: soft FTL is closed as a robust, calibratable, controllable, resource-bounded, material-embedded, repeatable hierarchical remote-scene protocol, and hard FTL is closed as an audited open-class, calibratable, controllable, resource-bounded, material-embedded, repeatable transport protocol for one-pole and composite sectors.

This is why the present chapter stops here. After E-C6, what remains is no longer the same kind of theoretical closure. What remains is engineering selection of concrete media, controls, parameter scales, and prototype architectures.

B Failure Modes and Non-Claims

This appendix rewrites the audit-facing negative blocks of the manuscript in companion form. Its role is not to improvise a generic warning chapter, but to preserve the exact audit logic that the manuscript repeatedly enforces: a strong theory must expose not only constructions and theorem packages, but also the ways those packages fail, the conditions under which projections cease to be admissible, and the claims that the current formal status does not authorize.

The manuscript is especially severe about this in the FTL program. Because observer-relative and spectral statements often sound stronger than they are, every gain in theorem power must be paired with an equally explicit negative mirror. The present chapter therefore follows the audit order already implicit in the hard-closure sequence and the final audit boundary: local route failures first, closure-boundary failures second, and forbidden inferences last.

B.1 Soft-Branch Failure Modes

B.1.1 Loss of readable detector contrast

The manuscript states the first soft failure already at the earliest witness level. Binary and finite-alphabet witness structures depend on positive operational contrast. If perturbations, classicalization error, memory distortion, or statistical load raise the effective operational load above the available contrast margin, then the witness ceases to be readable. In hierarchical language this becomes the scale-wise failure condition

$$R_\ell \geq \Delta_{\min}^{(\ell)}.$$

Then the corresponding scale is no longer operationally distinguishable.

This is not a minor technicality. The later scanner and hierarchy theorems inherit the binary witness logic rather than replace it. So when the contrast inequality fails, the later scanner package also fails at that scale.

B.1.2 Loss of hierarchy consistency

The manuscript also stresses that a hierarchical scanner is not a bag of unrelated labels. If finer labels no longer refine coarser ones, the observer does not reconstruct a coherent remote scene but

only a collection of inconsistent local records. This is one of the most important soft non-claims: a family of readable labels is not yet a remote scene unless hierarchical compatibility is preserved.

B.1.3 Premature unselective averaging

Another source-level failure occurs when one moves too quickly from selective dynamics to unselective averaging. Selective dynamics preserve branch-level information that unselective evolution can wash out. Therefore a protocol that exists only as a selective conditioned scan must not be advertised as an unconditional remote-imaging architecture. The manuscript closes the soft branch only as a branch-locked, memory-persistent remote-scene approximation package under its own detector and memory assumptions.

B.1.4 Memory-persistence failure

The manuscript then adds the memory barrier. If the memory lifetime is shorter than the sum of readout, write, and decode times,

$$\tau_{\mathrm{mem}}^{(\ell)} \leq t_{\mathrm{read}}^{(\ell)} + t_{\mathrm{write}}^{(\ell)} + t_{\mathrm{dec}}^{(\ell)},$$

then a readable detector shadow does not mature into a persistent local record. In that case the soft route remains a transient witness rather than an operational scanner.

B.1.5 Repeatability failure

In the late pre-engineering source, soft failure acquires a repeatability form. If accumulated reset, medium, or statistical drift drives

$$R_{\mathrm{soft,eff}}^{(\ell)}(m) \geq \Delta_{\mathrm{min}}^{(\ell)},$$

then the branch is no longer repeatable through m cycles even if one cycle had been feasible. This is the first place where a soft package can be theorem-closed and yet still fail as a reusable protocol.

B.2 Hard-Branch Failure Modes

B.2.1 Carrier-window collapse

The first hard failure is loss of the superluminal carrier margin. If

$$M_{\mathrm{vel}} \leq 0,$$

then the selected transported branch no longer outruns the chosen luminal reference sector and the regime is not hard FTL in the observer-relative sense of the manuscript.

B.2.2 Gap collapse

The next hard failure is loss of simple-pole separation. If

$$M_{\mathrm{gap}} \leq 0,$$

or equivalently the avoided-crossing gap collapses on the relevant transport window, then the selected transported pole can collide with another branch. In that case the branch may remain spectrally suggestive while no longer supporting the hard one-pole carrier language that the theorem package requires.

B.2.3 Residue collapse

The manuscript also insists on the distinction between spectral branch and observable branch. Even when the transported branch exists internally, the projected residue may vanish. Then one has an internal fast branch without a physically observable transported sector. This is exactly why the manuscript inserted the three-resolvent architecture and refused to identify internal branch existence with physical hard transport.

B.2.4 Identity failure

Pole survival is not identity preservation. Large nonadiabatic leakage, large residue drift, large observable drift, or large observer-shadow drift can all lead to

$$\epsilon_{\mathrm{id}}^{\mathrm{cyc}} \not\ll 1.$$

In that case the returned object-sector is not the same physical one-pole object, even if a projected pole survived during the carrier phase. The manuscript repeatedly treats this as one of the decisive negative checkpoints of the hard branch.

B.2.5 Composite binding collapse

For bound transport, positivity of the carrier-phase binding margin,

$$B_T(Q, \lambda) > 0,$$

is essential. If $B_T \leq 0$, or if the composite reduced resolvent loses its isolated pole or its narrow long-lived resonance character, then no bound-object transport has been established even if constituent one-pole carrier phases exist individually. The manuscript explicitly blocks the inference

$$\text{constituent hard transport} \Longrightarrow \text{bound-object hard transport}.$$

That implication is false without composite binding observability.

B.2.6 Projection-class instability

The current hard robustness and observability theorems all assume that the observer regime and the physical projector class remain inside the same audited admissible type. If a perturbation changes the superselection or admissibility type of the projection channel, the present theorem package may simply stop applying. The manuscript therefore refuses to advertise universal robustness under arbitrary observer changes.

B.2.7 Repeatability failure

In the late pre-engineering blocks the hard branch acquires its own repeatability failure form. If cumulative drift terms satisfy

$$M_{\mathrm{vel}} - \delta_{\mathrm{vel}}(m) \leq 0, \quad M_Z - \delta_Z(m) \leq 0, \quad M_{\mathrm{bind}} - \delta_{\mathrm{bind}}(m) \leq 0,$$

or if the effective one-pole or composite identity losses exceed their calibrated budgets, then the hard package may remain valid for one cycle but fail as an n-cycle transport protocol.

B.3 Audit-Level Non-Claims

B.3.1 Soft branch does not prove unrestricted remote vision

The manuscript repeatedly warns that soft closure is not the same as arbitrary full-resolution remote vision. The actual theorem package is a robust hierarchical remote-scene approximation under explicit detector, memory, and robustness budgets. That result is already strong, but it is not equivalent to unconstrained image reconstruction.

B.3.2 Hard branch does not prove unconstrained macroscopic transport

The hard branch is formulated for one-pole object-sectors and then for composite bound sectors under positive binding margins and explicit observability conditions. Ordinary-language references to "moving objects faster than light" must therefore always be read through that sector-based restriction. The manuscript explicitly forbids jumping from one-pole or composite carrier theorems to unqualified macroscopic transport rhetoric.

B.3.3 Internal existence is not physical observability

The current FTL package does not collapse internal spectral existence, sector-resolved observability, and physical projected observability into one rhetorical shell. That collapse is blocked by design. Whenever the manuscript says that a branch or sector exists, one must still ask: internal branch, sector-resolved projected pole, or physical projected pole? The program does not authorize treating those levels as interchangeable.

B.3.4 Theorem closure is not engineering realization

Even after theorem closure, calibration, control closure, resource closure, material embedding, and repeatability, the result is still a pre-engineering theorem package. It identifies admissible medium classes, protocol inequalities, reset or refresh conditions, and control windows. It does not by itself deliver a built architecture.

B.3.5 Single-cycle feasibility is not repeatability

The manuscript isolates repeatability as its own final pre-engineering block for a reason. Reset, refresh, medium fatigue, and cumulative drift can break a protocol that works once. So a single successful cycle does not automatically imply repeatable protocol operation.

B.4 Final Audit Boundary

B.4.1 What H-C5 changes and what it does not change

After the hard robustness lifting theorem H-C5 of X.32.7, the remaining failures are no longer foundational gaps in the formal program. They are model-level or realization-level failures. The manuscript names the most important ones explicitly:

1. an admissible class may exist formally while no physically selected subclass is realized in the actual low-energy world;
2. the admissible class may be too narrow to support realistic carrier media;
3. composite hard transport may exist only for very special bound sectors;

4. observer-level closure may hold internally while later empirical normalization or material embedding fails.

These are not theorem-chain failures. They are downstream realization failures.

B.4.2 Programmatic honesty

The manuscript summarizes its audit discipline in the strongest possible way: failure modes are not embarrassing leftovers, but part of the theorem package. The manuscript becomes stronger, not weaker, when it says exactly where the soft and hard branches stop, exactly what they require, and exactly what they do not yet authorize.

This is why the present chapter acts as the negative mirror of the rest of the manuscript. The soft and hard constructions remain scientifically meaningful only so long as their margins, admissibility classes, and observer channels remain intact. The moment those supports are lost, the correct action is not to relabel the failure as success, but to name the failure precisely. That is the audit standard of the manuscript itself, and this manuscript preserves it directly.

C Theorem Ledger and Pre-Engineering Transition

This appendix records the theorem ledger of the manuscript and the transition from theorem closure to pre-engineering work. Its role is to separate three layers cleanly: imported semantics, proved route-level results, and the calibrated realization work that begins only after theorem closure is complete.

C.1 The Theorem Ledger

C.1.1 The hard closure order

Before the manuscript enters pre-engineering closure, it records the hard closure order

$$H\text{-}C1 \to H\text{-}C2 \to H\text{-}C3 \to H\text{-}C4 \to H\text{-}C5.$$

This is the order in which general carrier existence, projected-physical observability, identity preservation, composite binding, observability, and identity compatibility, and robustness lifting close the hard route. These general theorems are proved from the perturbative lemmas, positivity margins, and projection-compatibility hypotheses fixed in the hard chapter.

The order is strict. Carrier existence comes first. Projected-physical observability depends on the carrier layer. Identity preservation depends on projected observability. Composite observability depends on identity control of the constituent transport cycle. Robustness lifting comes last, because it extends the already closed one-pole and composite theorems to an admissible open class.

C.1.2 What is proved in the manuscript

At theorem level the manuscript contains four layers.

The imported layer fixes observer-relative semantics, luminal comparison, and admissibility.

The proved soft layer contains candidate selection, branch-locked witness structure, explicit and kernel-generated binary families, finite and hierarchical scanner theorems, detector-memory operationalization, and soft robustness.

The proved hard layer contains carrier selection, explicit one-pole transport, projected-pole survival for the explicit family, identity-preserving carrier cycles, composite binding criteria, explicit

bound transport, quantitative robustness margins, the five perturbative closure lemmas, and the general hard-closure theorems H-C1 through H-C5.

The integrated layer contains the minimal soft-hard composition law, mission admissibility, and local integrated control.

C.2 The Pre-Engineering Transition

After theorem closure the next dependency chain is

$$E\text{-}C1 \to E\text{-}C2 \to E\text{-}C3 \to E\text{-}C4 \to E\text{-}C5 \to E\text{-}C6.$$

These blocks are, in order: quantitative soft calibration, quantitative hard calibration, control closure, resource and thermodynamic closure, material embedding, and repeatability together with protocol composition.

The order is rigid. Control closure depends on calibrated feasibility regions. Resource closure depends on calibration and control. Material embedding depends on all previous layers. Repeatability depends on the earlier closures and therefore comes last.

C.3 Formalization Note

The definitions and theorem interfaces of Sections 3–5 are stable enough for downstream formalization and computation. In particular, the manuscript freezes the observer-relative semantics, the soft scanner hierarchy, the hard carrier families, the perturbation norm and closure margins used in H-C1 through H-C5, the integrated mission record, and the pre-engineering calibration quantities used in the companion material.

C.4 What Remains Downstream Rather Than Theorem Work

What remains beyond the present manuscript is of two kinds.

First, within the theory text itself, the hard route is closed under the explicit admissibility, perturbative, and projection-compatibility hypotheses fixed in Sections 4 and 6. The present manuscript therefore does not leave a missing route-level theorem frontier of the same type as the one already resolved in the explicit and general hard closure chain.

Second, beyond theorem closure proper, the downstream work is calibrated realization work: control implementation, resource budgeting, material selection, repeatability engineering, and prototype design.

The purpose of this appendix is precisely to mark that boundary. The theorem package is closed under the stated hypotheses of the manuscript; the downstream work is realization work rather than reconstruction of a missing hard frontier.

D References

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